



Complete note OF IS - Information System

Management Information System (Tribhuvan Vishwavidalaya)



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INFORMATION SYSTEM NOTE

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CHAPTER 1: INFORMATION SYSTEM

Data: Raw facts such as an employee's name and number of hours worked in a week, inventory part numbers or sales orders.

Information: A collection of facts organized in such a way that they have additional value beyond the value of the facts themselves.

MIS: A management information system (MIS) provides information that organizations require to manage themselves efficiently and effectively. Management information systems are typically computer systems used for managing.

THE FIVE PRIMARY COMPONENTS:

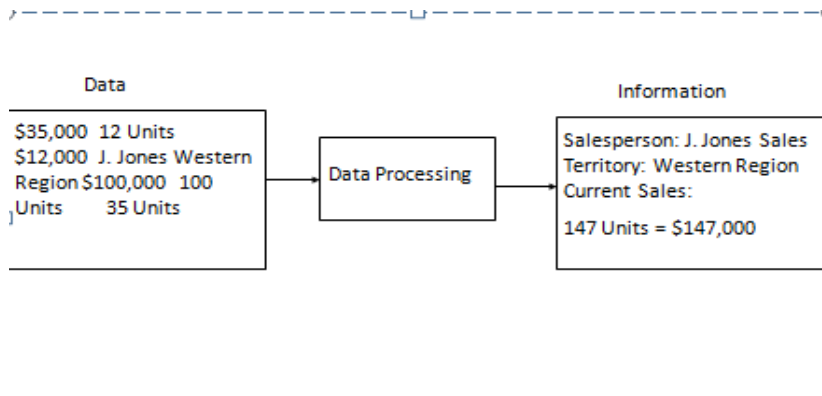
- 1.) Hardware, 2.) Software, 3.) Data (information for decision making), 4.) Procedures (design, development and documentation), and 5.) People (individuals, groups, or organizations).

MIS

Management information systems are distinct from other information systems because they are used to analyze and facilitate strategic and operational activities.

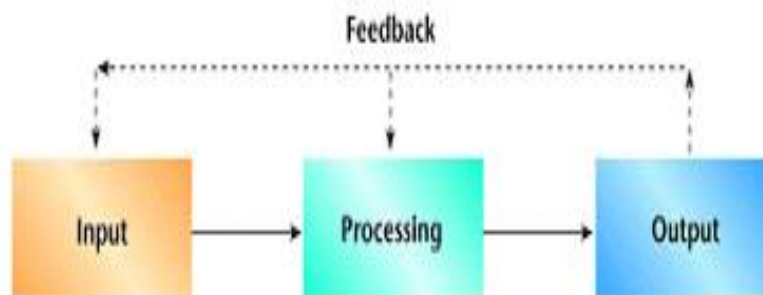
Academically, the term is commonly used to refer to the study of how individuals, groups, and organizations evaluate, design, implement, manage, and utilize systems to generate information to improve efficiency and effectiveness of decision making, including systems termed decision support systems, expert systems, and executive information systems.^[2] Most business schools (or colleges of business administration within universities) have an MIS department, alongside departments of accounting, finance, management, marketing, and sometimes others, and grant degrees (at undergrad, masters, and PhD levels) in MIS.

DATA VS INFORMATION

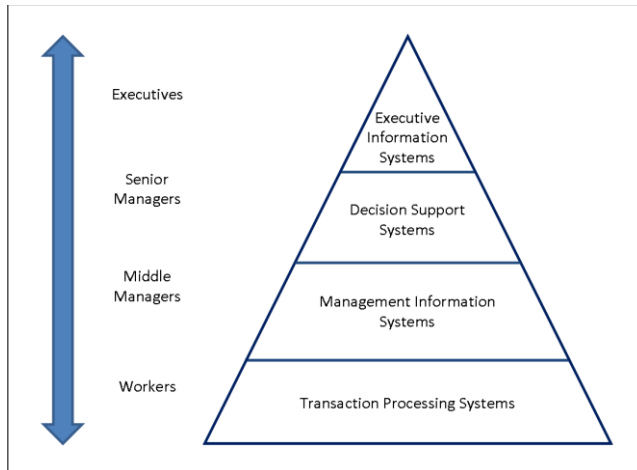


INFORMATION SYSTEM

- Information System is the study of complementary networks of hardware and software that people and organizations use to collect, filter, process, create, and distribute data.
- The IS's designer is concerned with how to use computer systems effectively in producing data for the right person at the right time.
- An information system is typically considered to be a set of interrelated elements or components that collect(input), manipulate(processes), and disseminate (output) data and information and provide a feedback mechanism to meet an objective.
 - Open System
 - Close System



TYPES OF INFORMATION SYSTEM



Transaction Processing Systems

A transaction processing system provides a way to collect, process, store, display modify or cancel transactions. Most of these systems allow multiple transactions to take place simultaneously. The data that this system collects is usually stored in databases which can be used to produce reports such as billing, wages, inventory summaries, manufacturing schedules, or check registers.

Management Information Systems

A management information system is an information system that uses the data collected by the transaction processing system and uses this data to create reports in a way that managers can use it to make routine business decisions in response to problems. Some of the reports that this information system creates are summary, exception and ad hoc reports. All this is done to increase the efficiency of managerial activity.

Decision Support Systems

A decision support system helps make decisions by working and analyzing data that can generate statistical projections and data models. This system gives support rather than replacing a managers judgement while improving the quality of a managers decision. A DSS helps solve problems while using external data.

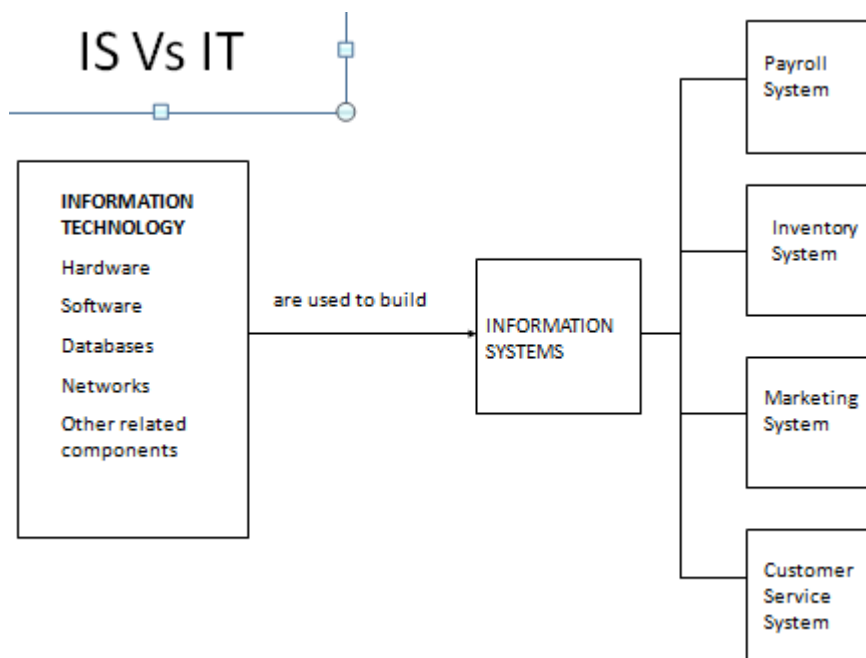
Executive Information Systems

Executive Information Systems are strategic-level information systems that are found at the top of the Pyramid. They help executives and senior managers analyze the environment in which the organization operates, to identify long-term trends, and to plan appropriate courses of action. The information in such systems is often weakly structured and comes from both internal and external sources. Executive Information System are designed to be operated directly by executives without the need for intermediaries and easily tailored to the preferences of the individual using them.

COMPUTER-BASED INFORMATION SYSTEM

- An Information System is an organized combination of people, hardware, software, communication networks and the data resources that collects, transforms and disseminates information in a organization.

INFORMATION SYSTEMS VS INFORMATION TECHNOLOGY



Information technology can be considered as a subset of information systems. It deals with the technology part of any information system, and as such deals with hardware, servers, operating systems and software etc.

- Information technology can be considered as a subset of information systems. It deals with the technology part of any information system, and as such deals with hardware, servers, operating systems and software etc.
- A system is always a combination of people, machines, processes and technology. And IT is just a part of the system. Since a part can never be identical to whole, information systems is never going to be identical to information technology. Designing of a system takes much more than technology as people and processes are also involved.

- 'Information systems' is in essence bridging the gap between business and the ever growing field of computers. On the other hand, information technology is all about managing technology and making use of it for the betterment of business.
- **Origin:** Information systems have been in existence since pre-mechanical era in form of books, drawings, etc. However, the origin of information technology is mostly associated with invention of computers.
- **Development:** Information systems have undergone great deal of evolution, i.e. from manual record keeping to the current cloud storage system. Similarly, information technology is seeing constant changes with evermore faster processor and constantly shrinking size of storage devices.
- **Business Application:** Businesses have been using information systems for example in form of manual books of accounts to modern TALLY. The mode of communication has also gone under big change, for example, from a letter to email. Information technology has helped drive efficiency across organization with improved productivity and precision manufacturing.

EXPANDING ROLES OF IS(HISTORY OF IS)

- Data Processing: 1950s-1960s
- Management Reporting: 1960s-1970s
- Decision support: 1970s-1980s
- Strategic and End User Support: 1980s-1990s
- Global Internetworking: 1990s-2000s
- **Data Processing: 1950s-1960s**
The first business application of computers (in the mid- 1950s) performed repetitive, high-volume, transaction-computing tasks. The computers "crunched numbers" summarizing and organizing transactions and data in the accounting, finance, and human resources areas. Such systems are generally called transaction processing systems (TPSs)
- **Management Reporting: 1960s-1970s**
Management Information Systems (MISs): these systems access, organize, summarize and display information for supporting routine decision making in the functional areas. Office Automation Systems(OASs): such as word processing systems were developed to support office and clerical workers.

- **Decision support: 1970s-1980s**

Decision Support Systems: were developed to provide computer based support for complex, non-routine decision.

- **Strategic and End User Support: 1980s-1990s**

The use or development of information systems by the principal users of the systems' outputs, such as analysts, managers, and other professionals.

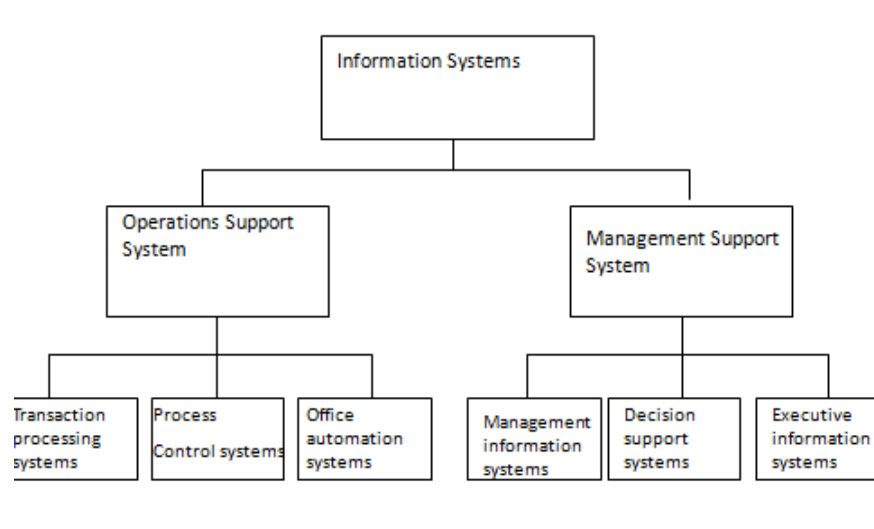
Intelligent Support System (ISSs): Include expert systems which provide the stored knowledge of experts to non-experts, and a new type of intelligent system with machine-learning capabilities that can learn from historical cases.

Knowledge Management Systems: Support the creating, gathering, organizing, integrating and disseminating of organizational knowledge.

- **Global Internetworking: 1990s-2000s**

Mobile Computing: Information systems that support employees who are working with customers or business partners outside the physical boundaries of their company; can be done over wire or wireless networks.

CLASSIFICATION OF IS



Operations support systems process data generated by business operations

- Major categories are:
Transaction processing systems
Process control systems
Office automation systems

Management Support Systems provide information and support needed for effective decision making by managers

- Major categories are:
- Management Information System
- Decision Support Systems
- Executive Information System
- 1. Operations Support System
- Transaction processing systems
- Process business exchanges
- Maintain records about the exchanges
- Handle routine, yet critical, tasks
- Perform simple calculations

Process control systems monitor and control industrial processes.

Office automation systems automate office procedures and enhance office communications and productivity.

Management support systems provide information and support needed for effective decision making by managers

Major categories are:

Management information systems

- Routine information for routine decisions
- Operational efficiency
- Use transaction data as main input
- Databases integrate MIS in different functional areas
- Some common examples of MIS output are reports on sales, stock, inventory, payroll

Decision Support System

- Interactive support for non-routine decisions or problem

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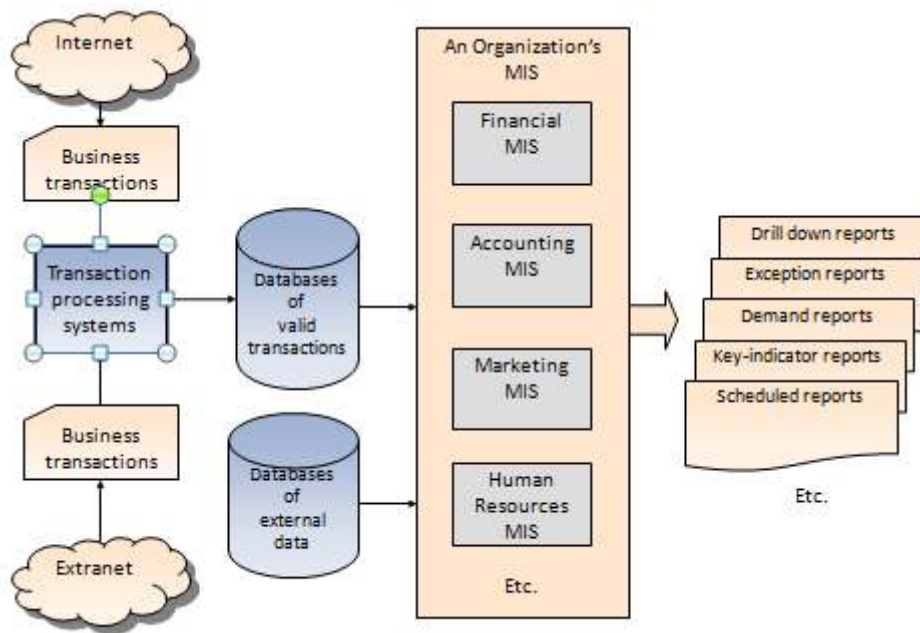
- End-users are more involved in creating a DSS than an MIS

Executive information systems

- provide critical information tailored to the information needs of executives
- DSS: Decision Support System

MIS: MANAGEMENT INFORMATION SYSTEM

- Functional Aspects
- MIS is an integrated collection of functional information systems, each supporting particular functional areas.



- **Financial MIS**

Provides financial information to all financial managers within an organization.

- **Marketing MIS**

Supports managerial activities in product development, distribution, pricing decisions, and promotional effectiveness

A FRAMEWORK FOR INFORMATION SYSTEMS ARCHITECTURE

- What is an Information Systems Architecture?

- An information systems architecture provides a unifying framework into which various people with different perspectives can organize and view the fundamental building blocks of information systems.
- Stakeholders have different views of the system and each has something “at stake” in determining the success of the system.
- Stakeholders can be broadly classified into four groups:
 - System Owners
 - System Users
 - System Designers
 - System Builders
- Information systems architecture provides a foundation for organizing the various components of any information system you care to develop.
- Different people have different views of the system. Managers, users, and technical specialists each view the system in different ways, and in different levels of detail. We call these people stakeholders in the system. They can be broadly classified into four groups:
 - System owners pay for the system to be built and maintained. They own the system, set priorities for the system, and determine policies for its use. In some cases, system owners may also be system users.
 - System users are the people who actually use the system to perform or support the work to be completed. In today’s team-oriented business world, system users frequently work side-by-side with system designers.
 - System designers are the technical specialists who design the system to meet the users requirements. In many cases, system designers may also be system builders.
- Systems builders are the technical specialists who construct, test, and deliver the system into operation.

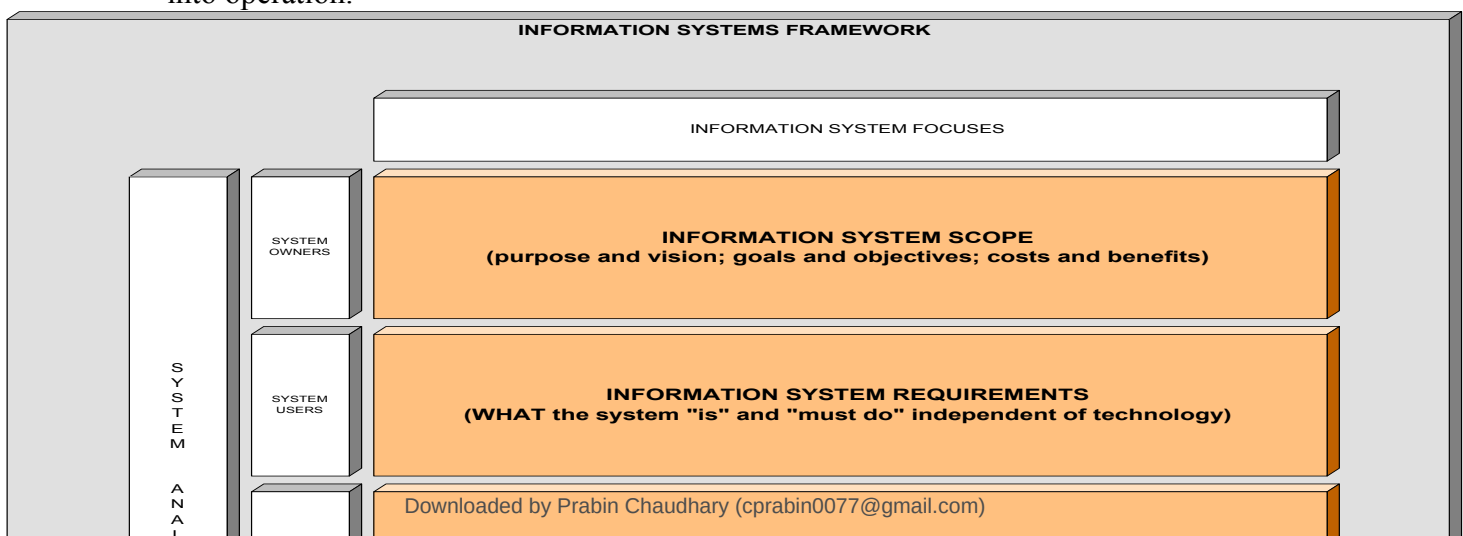


Figure :Information System Perspectives

- Each group of stakeholders is afforded one row in our information systems framework. Furthermore, each row has its own perspective or view of the information system.
- Perspectives - The People Side of Information Systems

- **What are Information Workers?**

- The term information worker (also called knowledge worker) was coined to describe those people whose jobs involve the creation, collection, processing, distribution, and use of information.

- **System Owners**

- System owners are an information system's sponsors and chief advocates. They are usually responsible for budgeting the money and time to develop, operate, and maintain the information system. They are also ultimately responsible for the system's justification and acceptance.
- All participants in the information systems game share one thing in common, they are what the U.S. Department of Labor now calls information workers.
- Today, more than 60 percent of the U.S. labor force is involved in the production, distribution, and usage of information.
- For any system, large or small, there will be one or more system owners.
- System owners usually come from the ranks of management. For medium-to-large information systems, the owners are usually middle or executive managers. For smaller systems, the owners may be middle managers or supervisors. For personal information systems, the owner and user are the same person.
- System owners tend to think in very general terms, not in details.

- System owners tend to be the least interested (or impressed) with the technology used in any information system. They are concerned with the ‘value’ returned by the system. Value is measured in different ways.
- What is the purpose of the system?
- What is the vision of the system – goals and objectives?
- How much will the system cost to build?
- How much will the system cost to operate?
- Will those costs be offset by measurable benefits?
- What about intangible benefits?

Perspectives - The People Side of Information Systems

- **System Users**

- System users are the people who use (and directly benefit from) the information system on a regular basis – capturing, validating, entering, responding to, storing, and exchanging data and information.
- There are many classes of system users including:
 - Internal Users
 - Clerical and service workers
 - Technical and professional staff
 - Knowledge workers are a subset of information workers whose responsibilities are based on a specialized body of knowledge.
 - Supervisors, middle managers, and executive managers
- System users make up the vast majority of the information workers in any information system.
- System users define (1) the problems to be solved, (2) the opportunities to be exploited, (3) the requirements to be fulfilled, and (4) the business constraints to be imposed by (or for) the information systems. They also tend to be concerned with how easy (or difficult) the system is to learn and use. Unlike system owners, system users tend to be less

concerned with costs and benefits of the system. Instead, they are concerned with 'business requirements' of the system.

- Internal users are employees of the business for which an information system is built. Internal users are the largest class of users and comprise the largest percentage of system users in most businesses.
- Clerical and service workers perform most of the day-to-day data processing in the average business. Most of the fundamental data in any business is captured or created by these workers, many of whom perform manual labors in addition to processing of data. Information systems that target these workers tend to focus on transaction processing speed and accuracy.
- Technical and professional staff consists largely of business and industrial specialists who perform highly skilled and specialized work. Their work is based on well-defined bodies of knowledge; hence, they are sometimes called knowledge workers. Information systems that target these knowledge works tend to focus on data analysis as well as generating timely information for problem solving.

- **Perspectives - The People Side of Information Systems**

- System Users
 - There are many classes of system users including: (continued)
 - Remote and Mobile Users
 - External Users
- System Designers
 - System designers translate users' business requirements and constraints into technical solutions. They design the computer files, databases, inputs, outputs, screens, networks, and programs that will meet the system users' requirements. They also integrate the technical solution back into the day-to-day business environment.
- Supervisors, middle managers, and executive managers are all decision makers. Supervisors tend to focus on day-to-day management issues. Middle managers are more concerned with tactical, or short-term management plans and problems. Executive managers are concerned with overall business performance, an strategic or long-term planning and problem solving. Information systems for management tend to focus entirely on information access. Managers need the right information at the right time to solve problems and make good decisions.

- Remote and mobile users like traditional internal users, they are employees of the business. Unlike traditional internal users, they are geographically separated from the business. An example is the sales and service representatives.
- Many business are looking to telecommuting to reduce costs and improve worker productivity. Telecommuting, stated simply, is working from home. There is considerable evidence to suggest that many employees can be just as productive working at home if they can be connected to the company's information systems through modern telecommunications technology.
- Businesses are redesigning their information systems to directly connect to and interoperate with their business and trading partners, suppliers, customers, and even the end consumer. The explosive growth of the Internet for electronic commerce is making the consumer a external user of information systems. Currently, World Wide Web pages on the Internet are mostly used to market information to the end consumer of products.

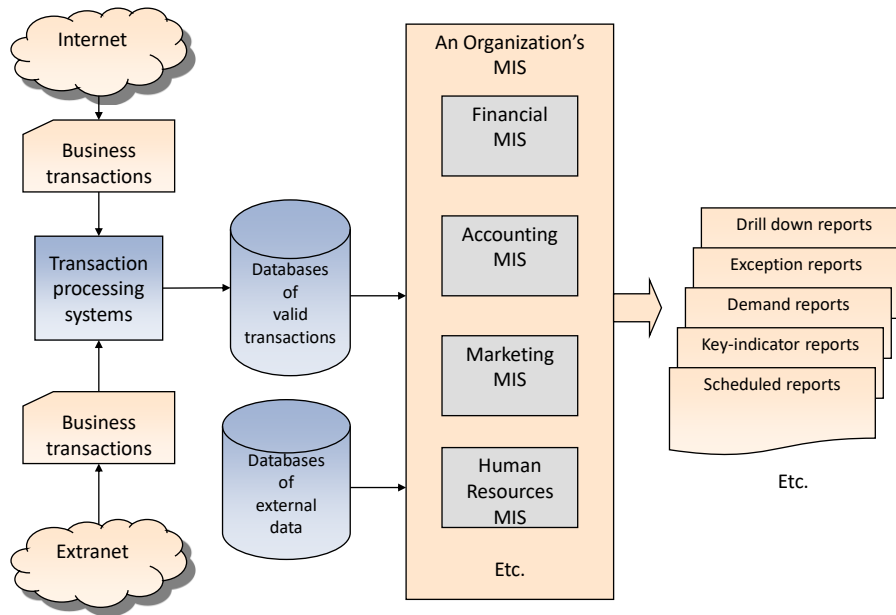
- **Perspectives - The People Side of Information Systems**

- System Builders
 - System builders construct the information system components based upon the design specifications from the system designers. In many cases, the system designer and builder for a component are one and the same.
 - The applications programmer is the classic example of a system builder.
- The Role of the System Analyst
 - For the system owners and users, the analyst typically constructs and validates their views.
 - For the system designers and builders, the analyst (at the very least) ensures that the technical views are consistent and compatible with the business views.
- Other technical specialists may also be involved, such as systems programmers, database programmers, network administrators, and microcomputer software specialists.

IS IN FUNCTIONAL AREA

MIS is an integrated collection of functional information systems, each supporting particular functional areas.

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QUALITIES OF INFORMATION SYSTEM

- Management information systems (MIS) is an organized approach to gathering information from company operations and making a strategic management decision. Developing quality characteristics for gathering information is essential to making solid management decisions.
- The main qualities of good management information system are:
 1. Relevance
 2. Accuracy
 3. Timely
 - 4. Exhaustive
 5. Cost-Effective

Main Qualities of Good Management Information System

- **Relevance:** Information should be relevant to the strategic decision that company management is currently reviewing. Because companies may review several business opportunities at one time, avoiding information not relating to the decision is essential.
- 2. **Accuracy:** MIS information should be accurate and avoid any probable costs. Making decisions based on estimates can lead to cost overruns or lower profits.

from future operations.

3. Timely: Many management decisions are based on information from a certain time period, such as quarterly or annual periods. Information outside of the requested time frame may skew information and lead to an improperly informed decision.

4. Exhaustive: MIS information gathering should resemble an upside-down triangle. The early stages of information gathering should be exhaustive, including all types of company information. As management narrows its decision-making process, the information is refined to include only the most relevant pieces.

5. Cost-Effective: The MIS needs to be a cost-effective and efficient system for gathering information. Most of these systems are developed internally, creating costs that cannot be passed to clients.

IS RESOURCES

- The individual data being processed through the use of hardware and software and shared through network connection has allowed us to utilize more information in less time.
 - Information Systems Resources
- Networks ...connected in some manner that allows to sharing of resources
- Hardware and Peripheral Devices ...tangible and can be touched
- Software ...intangible and can't be touched physically
- Data ...one piece of a record
- People ...work together to create usable information

CHAPTER 2

CONTROL, AUDIT AND SECURITY OF INFORMATION SYSTEM

CONTROLS

- Methods, policies, and procedures
- Ensures protection of organization's assets
- Ensures accuracy and reliability of records, and operational adherence to management standards

GENERAL CONTROLS

- **IT General Controls** – are policies and procedures that relate to many applications and support the effective functioning of application controls by helping to ensure the continued proper operation of information systems. These controls apply to mainframe, server, and end-user environments. General IT controls commonly include:
 - Controls over data centre and network operations
 - System software acquisition, change and maintenance
 - Access security
 - Application system acquisition, development, and maintenance.
 - Physical security of assets, including adequate safeguards such as secured facilities over access to assets and records,
 - Authorization for access to computer programs and data files.
- Establish framework for controlling design, security, and use of computer programs
- Include software, hardware, computer operations, data security, implementation, and administrative controls
 - **Application controls**
 - These are controls that relate to specific computer software applications and the individual transactions. For example, a company would usually place restrictions on which personnel have authorization to access its general ledger so as to revise its chart of accounts, posting / approving journal entries etc.
- Unique to each computerized application
- Include input, processing, and output controls

DIGITAL FIRM

- The Digital Firm is a general term for organizations that have enabled core business relationships with employees, customers, suppliers, and other external partners through digital networks.
- Protecting the Digital Firm
- High-availability computing: Tools and technologies enabling system to recover from a crash
- Disaster recovery plan: Runs business in event of computer outage
- Load balancing: Distributes large number of requests for access among multiple servers
- Mirroring: Duplicating all processes and transactions of server on backup server to prevent any interruption
- Clustering: Linking two computers together so that a second computer can act as a backup to the primary computer or speed up processing

AUDIT

- An audit is an objective examination and evaluation of the financial statements of an organization to make sure that the records are a fair and accurate representation of the transactions they claim to represent. It can be done internally by employees of the organization, or externally by an outside firm.
- Ensure computer based financial and other information reliable
- Ensure all records included while processing
- Ensure protection from frauds

Auditing Around Computer

- Take sample inputs and manually apply processing rules and compare outputs with computer outputs

Auditing Through The Computer

- Establish audit trail which allows examining selected intermediate results
- Control totals provide intermediate checks
- Facility to trace transaction value and print intermediate results
- Selective printing of records meeting criteria specified by the auditor

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- For example :Inactive accounts, overactive accounts, accounts with high balance
 - Comparing credit and debit balances
 - Ensure logs are kept of who did what in critical data entry and processing to fix responsibility. Called an Audit trail.
- Auditor's own check inputs and expected outputs.

Auditing With The Computer

- Extracting data based on the specified criterion for inspection(e.g. Students with wide disparity in marks in two subjects)
- Totaling specified subset of data for check
- Procedure to check sale discounts
- Process with independent data file created by auditor and verify to see if system is as per specification

SECURITY

- Security means protection of data from accidental or intentional modification, destruction or disclosure to unauthorized persons

POTENTIAL THREATS TO SECURITY

- Natural disasters such as fire, floods, earthquakes
- Accidents such as disk crashes, file erasure by inexperienced operators
- Theft/erasure of data by disgruntled employees
- Frauds by changing programs, data by employees
- Industrial espionage
- Viruses/Worms
- Hackers who break into systems connected to the internet
- Denial of service attacks by flooding with mail

HOW TO PROTECT DATA/PROGRAMS

- Regular back up of data bases every day/or week depending on the time criticality and size

- Incremental back up at shorter intervals
- Backup copies kept in safe remote location-particularly necessary for disaster recovery
- Duplicate systems run and all transactions mirrored if it is a very critical system and cannot tolerate any disruption before storing in disk.
- Physical locks
- Password system
- Biometric authentication (Eg: Finger print)
- Encrypting sensitive data/programs
- Identification of all persons who read or modify data and logging it in a file
- Training employees on data care/handling and security
- Antivirus software
- Firewall protection when connected to internet

ENTERPRISE LAYERED SECURITY STRATEGY

- practice of combining multiple mitigating security controls to protect resources and data.
- Also known as layered defense

Types of Security Layers

- Consumer Layered Security Strategy
- Enterprise Layered Security Strategy

CONSUMER LAYERED SECURITY STRATEGY

- Extended validation (EV) SSL certificates
- Multifactor authentication (also sometimes known as versatile or two-factor authentication)
- Single sign-on (SSO)
- Fraud detection and risk-based authentication
- Transaction signing and encryption
- Secure Web and e-mail
- Open fraud intelligence network

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Enterprise Layered Security Strategy

- Workstation application whitelisting
- Workstation system restore solution
- Workstation and network authentication
- File, disk and removable media encryption
- Remote access authentication
- Network folder encryption
- Secure boundary and end-to-end messaging
- Content control and policy-based encryption

WHAT IS E-COMMERCE SECURITY

- E-commerce security is the protection of e-commerce assets from unauthorized access, use, alteration, or destruction.
- **6 dimensions of e-commerce security**
 1. Integrity: prevention against unauthorized data modification
 2. Nonrepudiation: prevention against any one party from reneging on an agreement after the fact
 3. Authenticity: authentication of data source
 4. Confidentiality: protection against unauthorized data disclosure
 5. Privacy: provision of data control and disclosure
 6. Availability: prevention against data delays or removal

Threats: anyone with the capability, technology, opportunity, and intent to do harm. Potential threats can be foreign or domestic, internal or external, state-sponsored or a single rogue element. Terrorists, insiders, disgruntled employees, and hackers are included in this profile (President's Commission on Critical Infrastructure Protection)

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• Concern
• Loss of Privacy/confidentiality, data misuse/abuse
• Cracking, eavesdropping, spoofing, rootkits
• Viruses, Trojans, worms, hostile ActiveX and Java
• System unavailability, denial of service, natural disasters, power interruptions

1. Intellectual property threats -- use existing materials found on the Internet without the owner's permission, e.g., music downloading, domain name (cybersquatting), software pirating

2. Client computer threats

- Trojan horse
- Active contents
- Viruses

3. Communication channel threats

- Sniffer program
- Backdoor
- Spoofing
- Denial-of-service

4. Server threats

- Privilege setting
- Server Side Include (SSI), Common Gateway Interface (CGI)
- File transfer
- Spamming

COUNTERMEASURE

A procedure that recognizes, reduces, or eliminates a threat

1. Intellectual property protection

- Legislature
- Authentication

2. Client computer protection

- Privacy -- Cookie blockers; Anonymizer
- Digital certificate (Figure 5.9)
- Browser protection
- Antivirus software
- Computer forensics expert

3. Communication channel protection

- Encryption

* Public-key encryption (asymmetric) vs Private-key encryption (symmetric)
* Encryption standard: Data Encryption Standard (DES), Advanced Encryption Standard (AES)

- Protocol

* Secure Sockets Layer (SSL)

* Secure HyperText Transfer Protocol (S-HTTP)

- Digital signature

Bind the message originator with the exact contents of the message

—A hash function is used to transform messages into a 128-bit digest (message digest).

- The sender's private key is used to encrypt the message digest (digital signature)
- The message + signature are sent to the receiver
- The recipient uses the hash function to recalculate the message digest
- The sender's public key is used to decrypt the message digest
- Check to see if the recalculated message digest = decrypted message digest

4. Server protection

- Access control and authentication
 - * Digital signature from user
 - * Username and password
 - * Access control list
- Firewalls

International Computer Security Association's classification:

- Packet filter firewall: checks IP address of incoming packet and rejects anything that does not match the list of trusted addresses (prone to IP spoofing)
- Application level proxy server: examines the application used for each individual IP packet (e.g., HTTP, FTP) to verify its authenticity.
- Stateful packet inspection: examines all parts of the IP packet to determine whether or not to accept or reject the requested communication.

HOW TO MINIMIZE SECURITY THREATS

1. Perform a risk assessment à a list of information assets and their value to the firm
2. Develop a security policy à a written statement on:
 - * what assets to protect from whom?
 - * why these assets are being protected?
 - * who is responsible for what protection?
 - * which behaviors are acceptable and unacceptable?
3. Develop an implementation plan à a set of action steps to achieve security goals
4. Create a security organization à a unit to administer the security policy
5. Perform a security audit à a routine review of access logs and evaluation of security procedures

SSL CERTIFICATES

- SSL stands for Secure Socket Layer.
- It is the standard security technology for establishing an encrypted link between a web server and a browser. This link ensures that all data passed between the web server and browsers remain private and integral.
- To be able to create an SSL connection a web server requires an SSL Certificate.

- SSL Certificates are small data files that digitally bind a cryptographic key to an organization's details. When installed on a web server, it activates the padlock and the https protocol (over port 443) and allows secure connections from a web server to a browser. Typically, SSL is used to secure credit card transactions, data transfer and logins, and more recently is becoming the norm when securing browsing of social media sites.
- SSL Certificates bind together:
 - A domain name, server name or hostname.
 - An organizational identity (i.e. company name) and location.
- An organization needs to install the SSL Certificate onto its web server to initiate secure sessions with browsers.
- Depending on the type of SSL Certificate applied for, the organization will need to go through differing levels of vetting.
- Once installed, it is possible to connect to the website over `https://www.domain.com`, as this tells the server to establish a secure connection with the browser.
- Once a secure connection is established, all web traffic between the web server and the web browser will be secure. Browsers tell visitors a website is SSL secure via several visible trust indicators:



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Types of SSL Certificates

- Why there are different types of Certificates?
 - Some organizations need SSL simply for confidentiality, e.g. encryption
 - Some organizations wish to use SSL to enhance trust in their security and identity, e.g. they want to show customers they have been vetted and are a legitimate organization
- There are basically three types of SSL certificates
 - OV SSL CERTIFICATES: assures the validity of a Web site by verifying that the applicant is a legitimate business. Before issuing the SSL certificate, the CA performs a rigorous validation procedure, including checking the applicant's business credentials (such as the Articles of Incorporation) and verifying the accuracy of its physical and Web addresses.
 - DV SSL CERTIFICATES: The validation procedure is less rigorous for a Domain Validated SSL Certificate. When issuing a Domain Validated SSL Certificate, the CA checks only that the applicant's name and contact information matches the registration information in the WHOIS database for the domain name associated with the applied for SSL Certificate.
 - EV SSL CERTIFICATES: The Certificate application process itself is more thorough and the validation criteria more rigorous for EV certification, whose applicants, at least initially, are limited to certain types of business entities and government agencies.

EV (EXTENDED VALIDATION)SSL CERTIFICATES

Extended Validation, or EV SSL, raises the bar on standard SSL validation processes, incorporating some of the highest standards in identity assurance to establish the legitimacy of online entities.

- Certificate Authorities put applicant websites through rigorous evaluation procedures and meticulous documentation checks to confirm their authenticity and ownership.
- This systematic authentication process, also known as the Extended Validation Standard, is based on a set of guidelines prescribed for CAs to adhere to when they receive a request for a digital certificate from an organization or business entity.
- These guidelines include:
 - Establishing the legal, physical and operational existence of the entity
 - Verifying that the entity's identity matches official records like incorporation and business licensing information
 - Confirming that the entity owns or has exclusive rights to use the domain mentioned in the application for certification
 - Confirming that the request for an EV certificate has been authorized by the entity
- The objective of the EV issuance process is to enable users to distinguish legitimate websites from phishing sites, building their trust in online commercial transactions and increasing participation.



REMOTE BASED AUTHENTICATION

- Remote access is the ability to get access to a computer or a network from a remote distance through wired or wireless connection.
- Authentication is the method of proving the subjects identity. E.g.: Password, Passphrase, PIN

Why?

- To Prevent
- Accessing private data and information transferring between server and users i.e. Channel attack.

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- Direct attacks from hackers into network .
- Brute force, software attacks

Authentication Methods:

- Biometrics
- Passwords
- Cognitive Passwords
- Card Based
- One-Time or Dynamic Passwords (token based)

Importance Today ?

- Today every thing is electronics and internet based like e-banking, e-commerce, e-learning, e-governance, m-banking etc.
- Companies have many branches worldwide so data and information are distributed among branches offices.
- User do transaction remotely using internet using different handheld devices.
- All information of enterprises are centralized at server which is shared/distributed remotely among concerned people worldwide.

CONTENT CONTROL AND POLICY BASED ENCRYPTION

- Services for the security of email content in an organization
- Email content like:

credit card no., account information, etc.

- -organization vital information, customer
 - vital information.
- Email Content Control
- This service contains the rules that defines:
 - -whether an email is to be encrypted or not
 - - which block or header in email to be encrypted

Policy Based Encryption(PBE)

- It's a service that encrypts specific emails based on policy.
 - -Set of rules designed to analyze all email

- PBE uses the Email Content Control rules to identify which email needs to be encrypted.
- The PBE Service is managed through the same control panel that you use to manage your Anti-Virus.
- PBE service is closely integrated with the Email Content Control service.
- Policy Based Encryption Benefits:
 - Automatically applies email encryption based on the organization's email security policies.
 - Data loss prevention and email messages security policies are consistently and accurately applied.
 - Eliminates email encryption key management, backup and administration burdens.
 - -uses software-as-a-service (SaaS) infrastructure

CHAPTER 3: ENTERPRISE MANAGEMENT SYSTEM

ENTERPRISE MANAGEMENT SYSTEM

- EMS is concerned with control, monitoring and the management of IT infrastructure and applications in order to optimize IT service delivery in Company.
- EMS is wide information system designed to coordinate all the resources, information and activities needed to complete business processes.
- Enterprise – an entire company, everything, all-inclusive
- Management – The monitoring and controlling of entities
- Systems – Information Technology Infrastructure, hardware and software, data, information, and processes

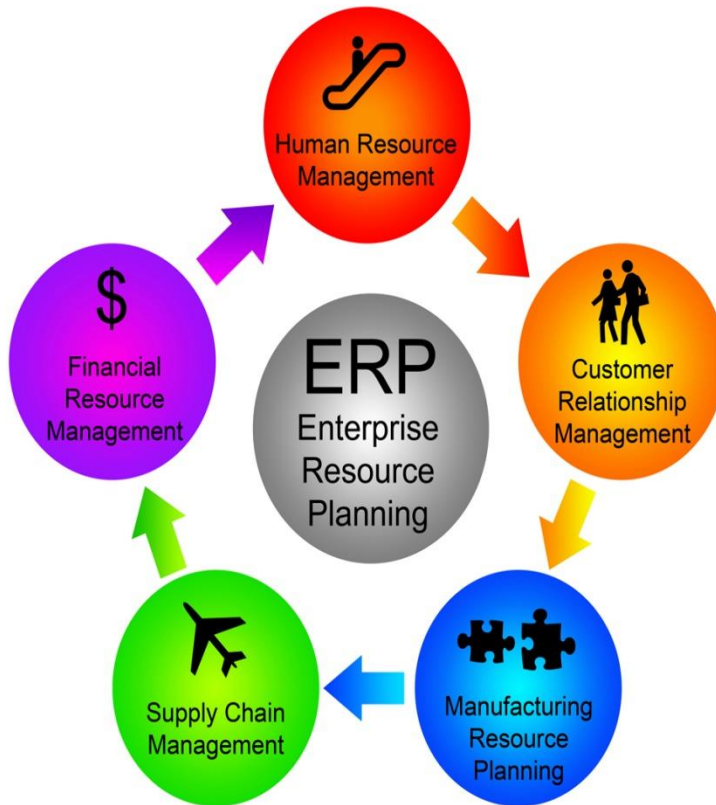
ENTERPRISE SOFTWARE

- Enterprise software is any software used in large organizations (whether business or government).
- It is considered to be an essential part of a computer-based information system, and it provides business-oriented tools such as online payment processing and automated billing systems.
- Enterprise software is also referred to as enterprise application software.

ENTERPRISE RESOURCE PLANNING

- ERP is business process management software that allows an organization to use a system of integrated applications to manage the business and automate back office functions.
- ERP software integrates all facets of an operation, including product planning, development, manufacturing processes, sales and marketing.
- Some of ERP's functions include:
 - Bookkeeping & Accounting
 - Human Resource Management
 - Planning Production
 - Supply Chain management

ERP COMPONENTS

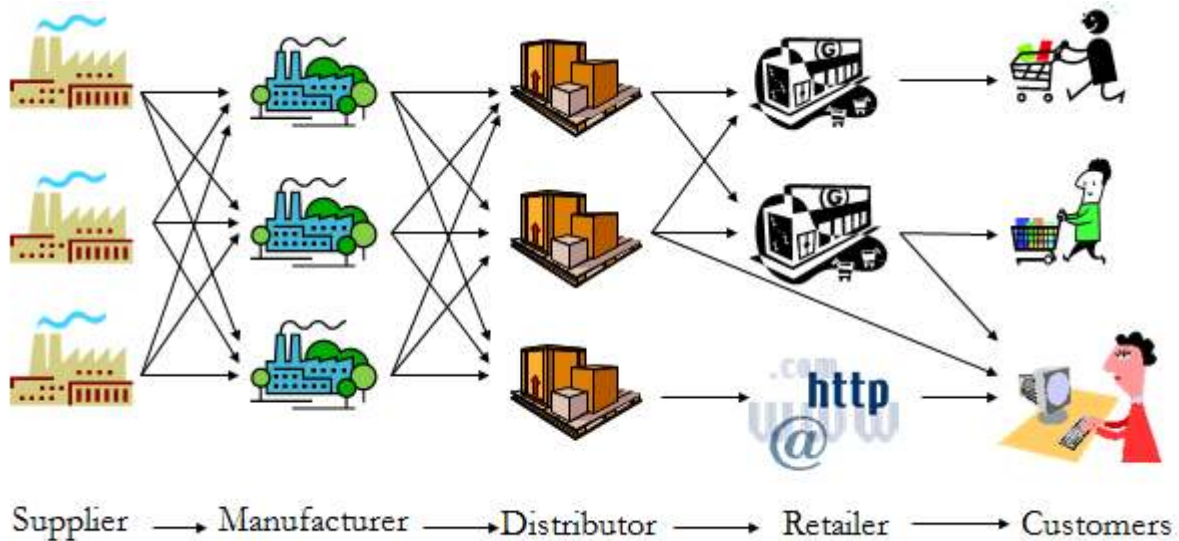


ENTERPRISE RESOURCE PLANNING SOFTWARE

- It helps an organization to integrate information flows, operations and processes all resources accessible, for example, materials, work force, machine and money.
- The focus of ERP is on resource management within constraints to maximize the return on investment.
- These data are then stored in a unified database, which are the key for the success of this software solution.
- The ERP package design is built on the principle of Best Practices.
- ERP Software: SAP, Supply Chain Management, CRM

SUPPLY CHAIN

- A supply chain is the system of organizations, people, activities, information and resources involved in moving a product or service from supplier to customer.
- Supply chain activities transform raw materials and components into a finished product that is delivered to the end customer.



Supply Chain Management

Supply Chain Management is

the design and management of processes
across organizational boundaries
with the goal of matching supply and demand
in the most cost effective way.

CUSTOMER RELATIONSHIP MANAGEMENT

- Customer Relationship Management is a strategy for managing all your company's interactions with current and prospective customers.
- CRM formation of bonds between a company and its customers.
- CRM enables your company to increase productivity, close more business, and improve customer satisfaction and retention.

CRM Strategies



Customer Retention Marketing Techniques

- Customization: Changing the product (not just the marketing message) according to user preferences
- Customer co-production: Allows the customer to interactively create the product
- Customer service tools include:
 - Frequently asked questions
 - Real-time customer service chat systems
 - Automated response systems

Benefits of using CRM

- Centralized customer interaction
- Improved customer support and satisfaction
- High rate of customer retention
- Increase revenue and referrals from existing customers
- Improve your products/services

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- Measure and optimize your performance
- Boost new business

ENTERPRISE INFORMATION MANAGEMENT

- Enterprise information management (EIM) is a set of business processes, disciplines and practices used to manage the information created from an organization's data.
- EIM initiatives seek to build efficient and agile data management operations with capabilities for information creation, capture, distribution and consumption.
- The goal is to provide and preserve information as a business asset that remains secure, easily accessible, meaningful, accurate and timely.

ENTERPRISE IT MANAGEMENT

- EITM is a strategy conceived and developed by Computer Associates International which details how organizations can transform the management of IT in order to maximize business value.
- Strategy for increasing the business relevance of the IT function, EITM considers the need for IT organizations to start operating as a service-based business.
- Ensuring investments are prioritized according to business strategy and that operational efficiencies can be more quickly realized and costs reduced when IT processes are integrated and automated.

ROLE OF IT IN ENTERPRISE MANAGEMENT

- Enterprise IT Management was developed in response to a growing need by IT organizations to gain more value from investments made in IT capabilities, infrastructure and resources.
- EITM proposes a set of capabilities that enable IT to better govern, manage and secure the IT services delivered to the business.
- IT/IS as asset, “strategic weapon”, “nervous system” (strategic level) Vs tool, commodity (operational level)

ENTERPRISE INFORMATION SYSTEMS

- Enterprise information system (EIS) is a system that serves an entire enterprise or at least two functional departments in:
 - Business intelligence (BI)
 - Enterprise resource planning (ERP)
 - Knowledge management (KM)
 - Partner relationship management (PLM)

- Business process management (BPM)
- Customer relationship management (CRM)

ROLE OF IS IN ENTERPRISE MANAGEMENT

- Help to unify the firm's structure and organization: One organization
- Management: Firm wide knowledge-based management processes
- Technology: Unified platform
- Business: More efficient operations & customer-driven business processes
- Supporting the major business functions: sales and marketing, manufacturing and production, finance and accounting, and human resources

ROLE OF IS AND IT IN ENTERPRISE MANAGEMENT

- Reduce Costs/ Improve Productivity
- Improve Customer Satisfaction/ Loyalty
- Create Competitive Advantage
- Generate growth
- Streamline Supply Chain
- Global Expansion

ENTERPRISE ENGINEERING

- Enterprise Engineering is integrated set of disciplines for building or changing an enterprise, its processes, and systems.
- It integrates the most powerful change methods and makes them succeed.
- The goal is a human-technological partnership of maximum efficiency in which learning takes place at every level.

Goal of the Enterprise Engineer

- Identify and integrate the most valuable and successful ways to change an enterprise, and to take them into a professional discipline with a teachable methodology and measures of effectiveness.

ENTERPRISE INTEGRATION

- Integration of markets
- Integration between several development and manufacturing sites

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- Integration between suppliers and manufacturers
- Integration of design and manufacturing
- Integration of multi-vendor hardware and software components

Basic principles for integration

- Provide the vision, right information, resources, and responsibility
- Empowered people
- A comprehensive and effective communication networks
- Democratization and dissemination of information
- Freely shared information

Types of Integration

- Loose Integration versus Full Integration
- Horizontal Integration versus Vertical Integration
- Intra-Enterprise Integration versus Inter-enterprise Integration
- System Integration, Application Integration, and Business Integration

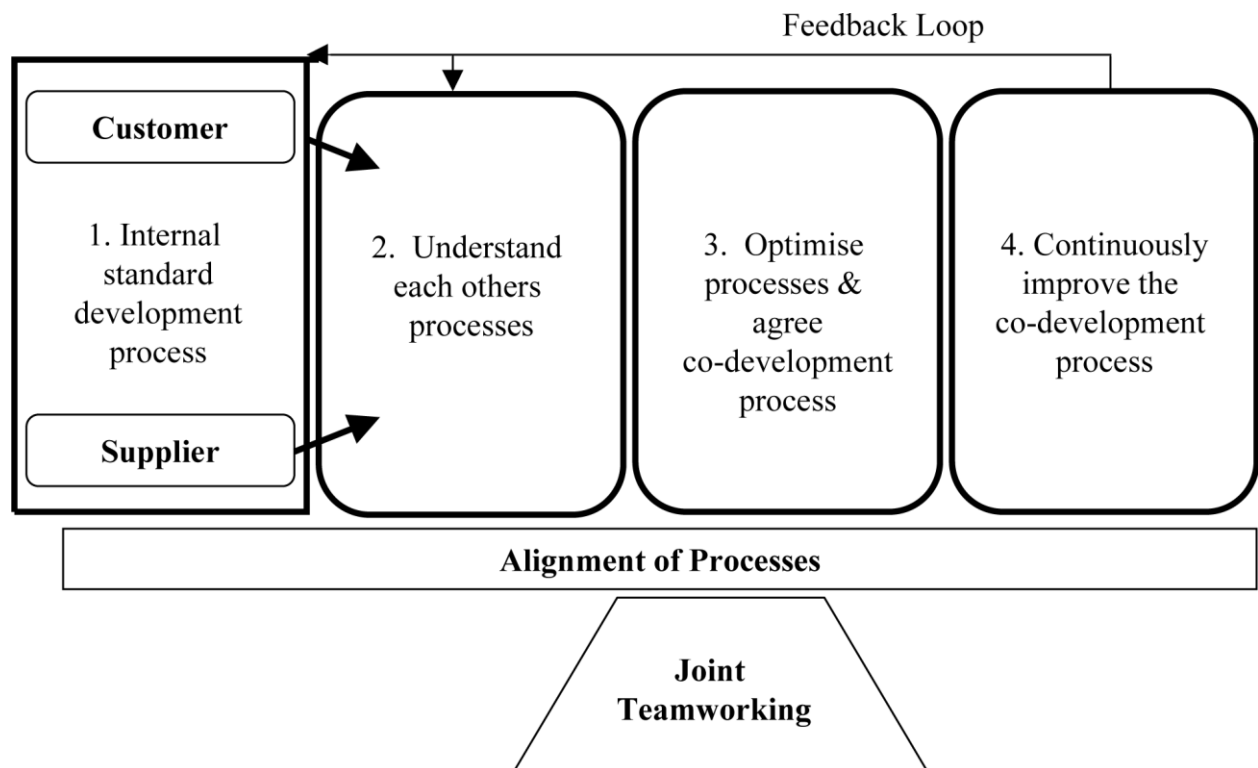
Loose Integration versus Full Integration

- Loose Integration - If two systems can merely exchange information with one another with no guarantee that they will interpret this information the same way
- Full integration - Two systems are fully integrated if and only if
 - the specificities of any one of these systems are only known to the system itself and not by the other one,
 - the two systems both contribute to a common task, and
 - the two systems share the same definition of each concept they exchange

ALIGNMENT PROCESS

- Developing a common understanding among the key stakeholders of the purpose and goals of the project and the means and methods of accomplishing those goals is called the Alignment Process.
- It is important to accomplish this alignment during the initiation phase.
- Project managers usually conduct a start-up meeting that is sometimes called a kickoff meeting.

- The agenda and duration of the start-up meeting depends on the complexity level of the project.
- Projects with a limited scope and short duration may engage in a session start-up meeting over lunch.
- A medium-complexity project will require more-hour meeting while a high-complexity project cannot achieve alignment in a single meeting. Alignment can require several days of activities.



Objective of Alignment Process

- The purpose of the alignment process is to develop a common understanding of the purpose, agree on the means and methods, and establish trust.
- The components of the alignment process are discussions of the purpose, goals, participant roles, methods of tracking progress and costs, methods of managing change, and building trust.
- The effects of a lack of trust are delays caused by fact checking or missing information that was not shared because the person's discretion was not trusted to handle sensitive information.

ELECTRONIC ORGANISMS

- As systems become more complex, the designs of this systems must be automated.
- Electronic organisms, in fact all organisms,

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- have to be complex, because they have to contain all the creative infrastructure necessary for their creation, reproduction, maintenance and action,
- but they can easily afford to be complex, because there is no need for detailed communication with a programmer.
- Electronic Organisms have the ability to react immediately to unforeseen challenges, without the need for a programmer to recognize the situation and deal with it by modifying a program.
- Electronic Organisms do so by recurrence to fundamental goals and organizing principles, just as programmers do so now.
- Electronic organisms will live, grow and evolve in the rapidly growing world of installed computers and networks, just as microbes, plants and animals live in natural ecosystems.
- Strong forces are pushing technology towards electronic organisms.

CHAPTER 4: DECISION SUPPORT AND INTELLIGENT SYSTEM

DECISIONS

- Information is used to make decisions. Decision making is not a single activity that takes place all at once.
- The process consists of several different activities that take place at different times.
- The decision maker has to identify and understand problems.

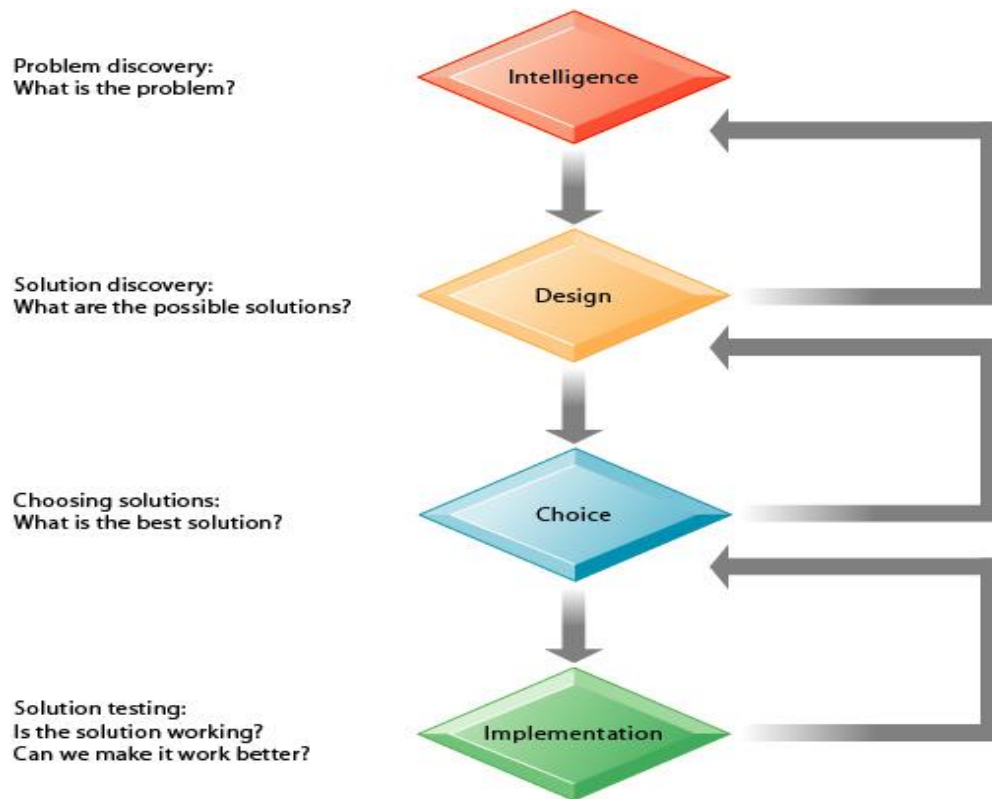
Four different stages in decision making are

- Intelligence
- Design
- Choice
- Implement

DECISION SUPPORT SYSTEM

- helps the information system in the intelligence phase to identify the problem and then go to the design phase for solution.
- The choice of selection criteria varies from problem to problem.
- It is required to go through these phase again and again till the satisfactory solution is found.

DECISION MAKING STAGES



DECISION SUPPORT SYSTEM

- is an integrated set of computer tool
- allows a decision maker to interact directly with computers to create information
- useful in making semi-structured and unstructured decisions.
- The software components for decision-support systems are a language system which enables the user to interact with the decision-support system, a problem-processing system.
- help decision makers and to identify and solve problems and make decisions.
- serve the management, operations, and planning levels of an organization and help to make decisions
- which may be rapidly changing and not easily specified in advance.
- DSS are used to collect data, analyze and shape the data that is collected, and make sound decisions
- The nature of the decision is such that the decision makers need a variety of information
- methods of decision making a change from time to time.

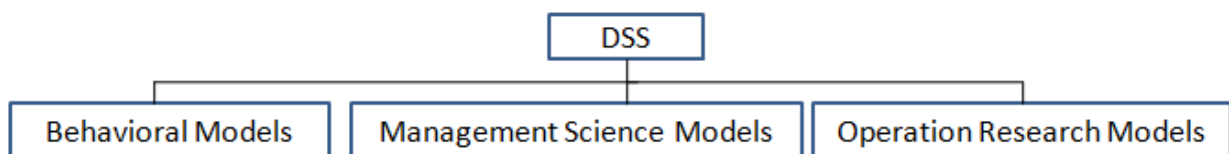
DSS TYPES OF DECISIONS

- Structured / Programmed Decisions :
 - Schedule decisions
 - Organization develops specific process for handling
 - Rules of decision making system are predetermined
- Unstructured / Non-programmed Decisions :
 - Repetitive decisions
 - Handled by general problem solving process
 - Decision taken by Decision Support Systems
 - Rules of decision making system are not fixed or predetermined
 - It requires every time the user has to go through the decision-making cycle.
 - Decision support systems can be built in case of programmable decision situation.
 - Semistructured Decisions:
 - Some decision procedures can be specified in advance, but not enough to lead to a definite recommended decision

DECISION SUPPORT SYSTEM MODELS

- Types of Decision Support System Models :
 - Behavioural Models
 - Management Science Models
 - Operation Research Models

Types of Models



Behavioural Models

- Behavioural Models :

- The decision maker can make the decisions for such behavioral relationships.

For eg: The trend (development) analysis, forecasting and statistical analysis models.

- The trend analysis indicates how different variables behave in trend setting in the past and hence in future.
 - In Market Research method, they can forecast or judge the behavior of the customers buying decisions. (i.e. The questionnaire are designed and computerized to evaluate customer's buying behavioral).

Management Science Models

– Management Science Models :

- These models are developed on the principles of the business management, accounting and economics.

For eg: the budgetary systems, cost accounting system, inventory management system.

- In the budgetary system, budgets are used for planning and control.
- In all the organization, budgets are prepared with the use of graphical representation in the form of line charts or bar charts.
- For eg : Sales Budget, Production Budget

Operation Research Models

– Operation Research Models :

- are the mathematical models in providing guidelines to managers for making effective decisions within the state of the current information
- Some applications
 - Telecommunications/Road/Rail Network Design
 - Organization Supply Chain Strategy
 - Revenue, Pricing and Promotions
 - Demand Forecasting
 - Project Planning

For eg : Linear Programming is mathematical modeling technique useful for guiding quantitative decisions in business planning, industrial engineering, and—to a lesser extent in the social and physical sciences.

BENEFITS OF DECISION SUPPORT SYSTEM

- Benefits of Decision Support System :
 - Ability to view data and sensing the problem through the different view.
 - Ability to understand and evaluate the business performance.
 - Ability to understand the problem and its result, and ability to judge the impact on business.
 - Ability to evaluate the impact of any change in the business performance and enabling to focus on the areas where impact is negative.
 - Ability to view the complex scenario or problem and to analyze it and develop alternatives to solve the problem.
 - Ability to make a better decisions due to quick analysis.
 - Ability to control the risk exposure in decisions.

GROUP DECISION SUPPORT SYSTEM

- Group Decision Support System
 - Information technology supports decision-making where there is a group participation. Such decision support system is called as Group Decision Support Systems (GDSS).
 - GDSS has come to mean computer software and hardware used to support group functions and processes.
 - GDSS supports Manager and Staff working in groups.
- There are four configurations of group members are possible.
 - Group members in one room operating on network with common display screen to share the display for all members.
 - Group members sit at their respective locations and use their desktop to interact with other members.
 - Group members are in different cities and they come together through teleconferencing or video conferencing with prior planning GDSS operations.
 - Group members are at remote locations may be in different countries and they come together through long distance telecommunication network.

WHY USE GDSS?

- High level managers can spend 80% of their time making decisions in groups. Applied correctly, GDSS can reduce this time, arriving at a better decision faster.
- GDSS provides the hardware, software, databases and procedures for effective decision making.

ADVANTAGES OF GDSS

- Parallel Communication – eliminate monopolizing, providing increased participation, better decisions
- Automated record keeping – no need to take notes, they're automatically recorded
- Ability for virtual meetings – only need hardware, software and people connected
- Portability - Can be set up to be portable...laptop
- Global Potential - People can be connected across the world.

DISADVANTAGES OF GDSS

- Cost – infrastructure costs to provide the hardware and software/room/network connectivity can be very expensive
- Security – especially true when companies rent the facilities for GDSS; also, the facilitator may be a lower level employee who may leak information to peers
- Technical Failure – power loss, loss of connectivity, relies heavily on bandwidth and LAN/WAN infrastructure properly setup system should minimize this risk

ENTERPRISE DECISION SUPPORT SYSTEM

- is a multi-disciplinary organization
- provides technical and program/project leadership for Departmental Information Technology, applications and systems.
- use decision support systems to quickly perform complex analyses over large amounts of data
- results are used to inform critical business decisions.



EXECUTIVE DECISION SUPPORT SYSTEM

- Executive Decision Support System is a computer-based technology
- designed specifically for the information needs of top executives level and provides for:
 - Rapid access to timely information;
 - Direct access to management reports;
 - Very user friendly and supported by graphics.
- Executive Decision Support Systems support the informational roles of executives.
- Executive Support Systems – specialized decision support systems designed to meet the needs of senior management.
- Exception reporting – reporting of only the results that deviate from a set of standards.
- Drill down reporting – investigating information in increasing detail.
- Include analysis support, communications, office automation and intelligence support.

KNOWLEDGE

- Knowledge is something that comes from information processed by using data.
- applied by knowledge workers who are involved in a particular job or task.

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- making decisions as well as many other actions
- Knowledge is information that is contextual, relevant, and actionable.

DATA, INFORMATION AND KNOWLEDGE

- Data refers to isolated facts such as individual measurement No meaning on their own
- Information
fact about situation, person, events
- Knowledge
Experience

KNOWLEDGE MANAGEMENT

- Knowledge Management (KM) comprises a range of strategies and practices
- used in an organization to identify, create, represent, distribute, and enable adoption of insights and experiences
- KM is a process that helps organizations identify, select, organize, disseminate, and transfer important information and expertise that are part of the organization's memory.

KNOWLEDGE MANAGEMENT SYSTEM

- Knowledge Management System refers to a (generally IT based) system for managing knowledge in organizations for supporting creation, capture, storage and distribution of information.
- Knowledge Management System
 - Create : Knowledge must be created either within or outside the organization.
 - Store : Knowledge can be stored somewhere, for others to find and use.
 - Find : Those who need the specific knowledge must then find out where it is, when they need it, by searching in the right places and asking the right people.
 - Acquire : Once the knowledge source is found, the user will then go through the act of actually acquiring it
 - Use : Once acquired, the knowledge can be put to use towards some productive purpose.
 - Learn : Having been used, perhaps repeatedly, the user will learn what worked well and not so well as a result of applying the knowledge gained.
- helps gather, organize, and share business knowledge within an organization

- databases that store and disseminate business knowledge. It may also be called knowledge bases.

KNOWLEDGE BASED EXPERT SYSTEM

- A Knowledge Based System is a computer program
- that uses artificial intelligence to solve problems within a specialized domain that ordinarily requires human expertise.
- involve classification, diagnosis, monitoring, design, scheduling, and planning for specialized tasks.
- Knowledge-based system is a more general than the expert system.

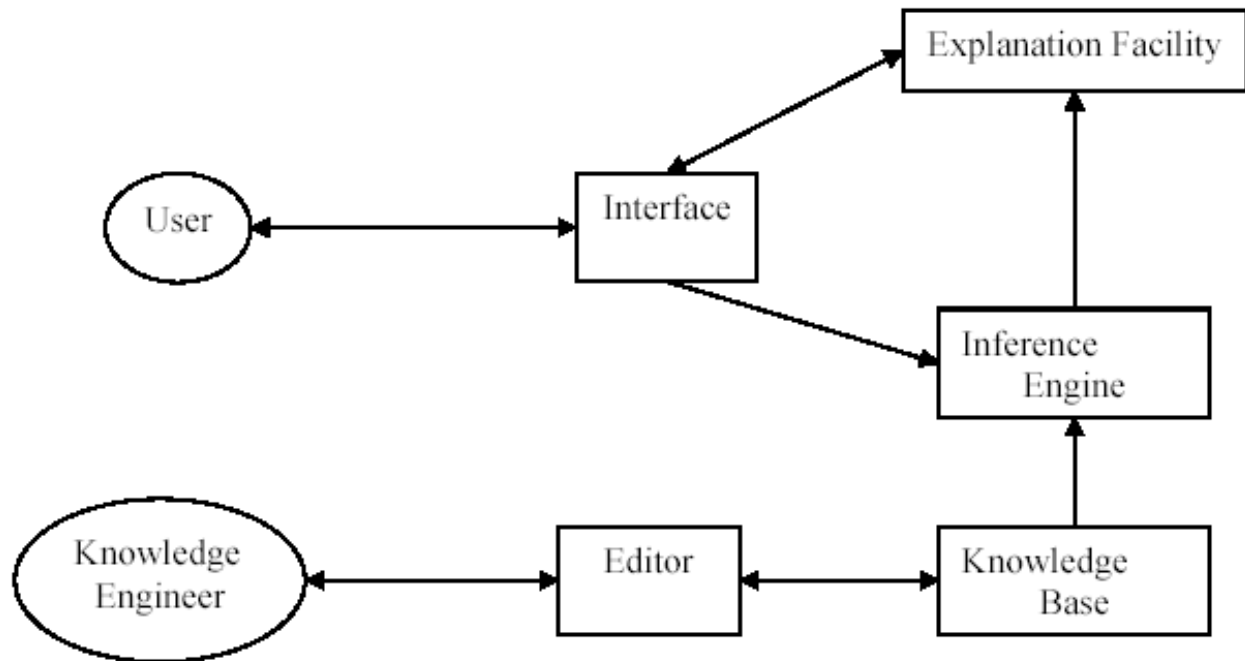
EXPERT SYSTEM

- Expert System is an extension of the decision support system.
- Expert system is an information system application that captures the knowledge and expertise of a problem solver or decision maker, and then simulates the ‘thinking’ of that expert for those who have less expertise.
- Expert systems are implemented with artificial intelligence technology, often called expert system shells.
- Expert System application areas in action

1. Medical Diagnosis

2. Telephone Network Maintenance

EXPERT SYSTEM ARCHITECTURE



INFERENCE ENGINE

- An inference engine tries to derive answers from a knowledge base.
- It is the brain of the expert systems that provides a methodology for reasoning about the information in the knowledge base, and for formulating conclusions.
- User Interface

It enables the user to communicate with an expert system.

KNOWLEDGE ENGINEER

- Knowledge Engineer
 - A knowledge engineer is a computer scientist who knows how to design and implement programs that incorporate artificial intelligence techniques.
- Knowledge Engineering
 - The art of designing and building the expert systems is known as Knowledge Engineering, knowledge engineers are its practitioners.
 - Knowledge engineering relies heavily on the study of human experts in order to develop intelligent & skilled programs.
 - The engineer then translates the knowledge into a computer- usable language,
 - and designs an inference engine, a reasoning structure, that uses the knowledge appropriately.

- He/she also determines how to integrate the use of uncertain knowledge in the reasoning process, and what kinds of explanation would be useful to the end user.
- When the expert system is implemented, it may be:
- An expert system is judged to be entirely successful when it operates on the level of a human expert.

CHARACTERISTICS OF EXPERT SYSTEM

- Like a human expert, an expert system is expected to
- Be specialist : know facts and procedural rules
- Use heuristics : interpolate from known facts
- Justify its conclusions : to establish credibility and confidence.
- The user can ask: be able to learn : be able to absorb new knowledge and apply it estimate the reliability of its answer.

BENEFITS OF EXPERT SYSTEM

- Benefits of Expert System
 - Increased output and productivity
 - Decreased decision making time
 - Increased process and product quality
 - Reduced downtime (machine failure detect and repair time)
 - Capture of scarce expertise
 - Easier equipment operation
 - Elimination of the need for expensive equipment
 - Operations in hazardous environments (no human required.)
 - Ability to work with incomplete or uncertain information
 - Knowledge transfer to remote locations
 - Enhancement of other information systems

INTELLIGENT SYSTEM

- Intelligent systems is a term that describes the various commercial applications of AI.
- Artificial Intelligence (AI) is a subfield of computer science concerned with:

- studying the thought processes of humans;
- recreating those processes via machines, such as computers and robots.
- Behavior by a machine that, if performed by a human being, would be considered intelligent.
- Turing test is a test for artificial intelligence, in which a human interviewer, conversing with both an unseen human being and an unseen computer, cannot determine which is which: named for British Mathematician AI pioneer (Alan Turing).

ARTIFICIAL INTELLIGENCE

- A field of science and technology based on disciplines such as computer science, biology, psychology, mathematics, & engineering
- At the machine level, think like human and act like human means Artificial Intelligence.
- Goal is to develop computers that can think, see, hear, walk, talk, and feel.
- Major thrust to development of computer functions normally associated with human intelligence reasoning, learning, problem solving

DOMAINS OF ARTIFICIAL INTELLIGENCE

- Domains of AI
 - Three major areas
 - Cognitive science
 - Robotics
 - Natural interfaces
- Cognitive science
 - Focuses on researching how the human brain works & how humans think and learn
 - Applications
 - Expert systems
 - Neural networks
 - Intelligent agents

DOMAINS OF ARTIFICIAL INTELLIGENCE

- Robotics

- Produces robot machines with computer intelligence and computer controlled, human like physical capabilities
- Natural interfaces
 - Natural language and speech recognition
 - Talking to a computer and having it understand
 - Virtual reality

NEURAL NETWORKS

- Neural networks is a system of programs and data structures that approximates the operation of the human brain.
- good at recognizing subtle(sensitive), hidden and newly emerging patterns within complex data
- interpreting incomplete inputs.

VIRTUAL REALITY

- Virtual reality is plainly speaking, seeing an imaginary world, rather than the real one.
- Seeing, hearing, smelling, testing, feeling.
- The imaginary world is a simulation running in a computer.
- The sense data is fed by some system to our brain.
- A medium composed of interactive computer simulations giving users the feeling of being present in the simulations.

Why Virtual Reality is needed?

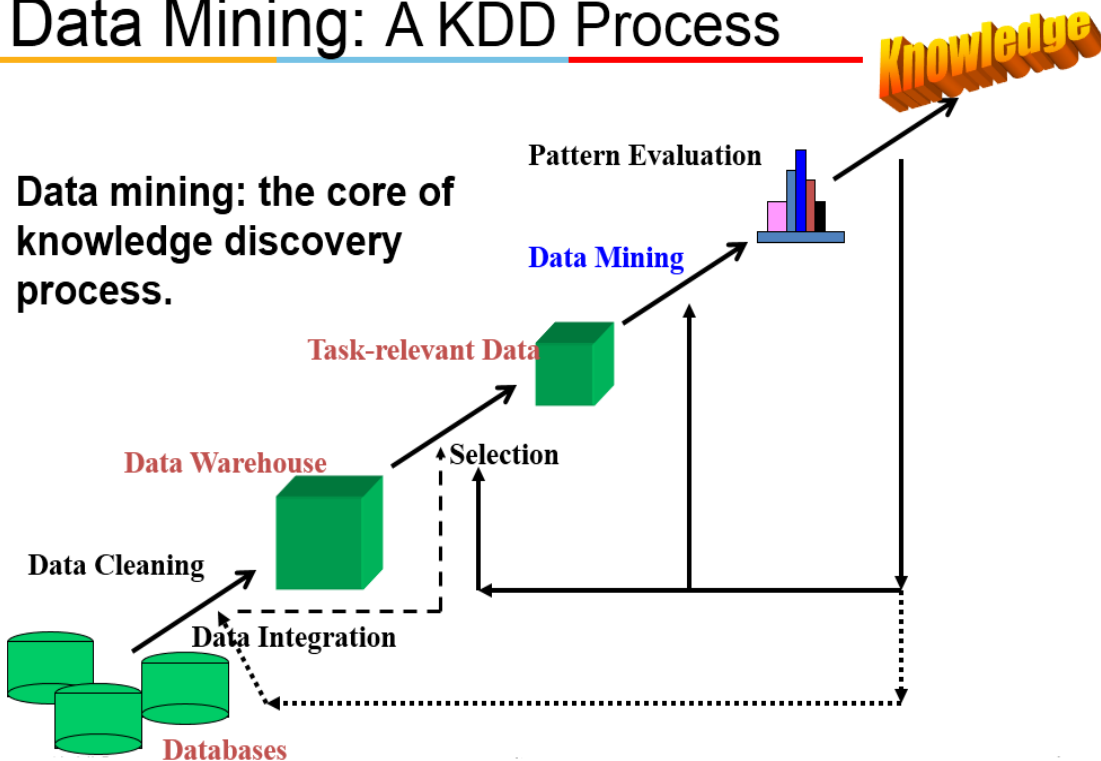
- Operations in dangerous environments
 - There are still many examples of people working in dangerous or hardship environments that could benefit from the use of VR-mediated teleportation.
 - Workers in radioactive, space, or toxic environments could be relocated to the safety of a VR environment where they could handle any hazardous materials without any real danger using tele operation or tele presence.
- Scientific Visualization
 - Scientific Visualization provides the researcher with immediate graphical feedback during the course of the computations and gives him/her the ability to 'steer' the solution process.

- Application at NASA Ames Research Center is the Virtual Planetary Exploration. It helps planetary geologists to remotely analyze the surface of a planet. They use VR techniques to roam planetary terrains.
- Medicine
 - Until now experimental research and education in medicine was mainly based on dissection and study of plastic models. Computerized 3D human models provide a new approach to research and education in medicine. Experimenting medical research with virtual patients will be a reality.
 - We will be able to create not only realistic looking virtual patients, but also histological and bone structures. With the simulation of the entire physiology of the human body.
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DATA MINING

- Data mining is a process used by companies to turn raw data into useful information.
- Data mining is the search for relationships and global patterns that exist in large databases but are 'hidden' among the vast amount of data
- Data Mining is also known as knowledge Discovery in Databases(KDD)
- Data mining refers to using a variety of techniques to identify nuggets of information or decision-making knowledge in bodies of data,
- and extracting these in such a way that they can be put to use in the areas such as decision support, prediction, forecasting and estimation.
- The data is often voluminous, but as it stands of low value as no direct use can be made of it; it is the hidden information in the data that is useful.

Data Mining: A KDD Process



STEPS OF A KDD PROCESS

- Learning the application domain:
 - relevant prior knowledge and goals of application
- Creating a target data set: data selection
- Data cleaning and preprocessing: (may take 60% of effort!)
- Data reduction and transformation:
 - Find useful features, dimensionality/variable reduction, invariant representation.
- Choosing functions of data mining
 - summarization, classification, regression, association, clustering.
- Choosing the mining algorithm(s)
- Data mining : search for patterns of interest
- Pattern evaluation and knowledge presentation
 - visualization, transformation, removing redundant patterns, etc.

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- Use of discovered knowledge

WHY IS DATA MINING NECESSARY?

- Make use of your data assets
- There is a big gap from stored data to knowledge; and the transition won't occur automatically.
- Many interesting things you want to find cannot be found using database queries

“find me people likely to buy my products”

“Who are likely to respond to my promotion”

- Credit ratings/targeted marketing:
 - Given a database of 100,000 names, which persons are the least likely to default on their credit cards?
 - Identify likely responders to sales promotions
- Fraud detection:
 - Which types of transactions are likely to be fraudulent, given the demographics and transactional history of a particular customer?
- Customer relationship management:
 - Which of my customers are likely to be the most loyal, and which are most likely to leave for a competitor?

DATA WAREHOUSE

- A data warehouse is a relational database that is designed for query and analysis rather than for transaction processing.
- Characteristics
 - Subject oriented, Integrated, Time Variant, Non-volatile
 - Web-based, Relational/multidimensional, Client/server, Real-time
 - Include metadata

Data Warehousing is process of constructing and using data warehouses which requires data integration, data cleaning, and data consolidation.

Data Warehouse—Subject-Oriented

- Organized around major subjects, such as customer, product, sales

- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process
- Constructed by integrating multiple, heterogeneous data sources
 - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
 - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - E.g., Hotel price: currency, tax, breakfast covered, etc.

When data is moved to the warehouse, it is converted.

Data Warehouse—Integrated

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Data Warehouse—Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems
 - Operational database: current value data
 - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
 - Contains an element of time, explicitly or implicitly
 - But the key of operational data may or may not contain “time element”

Data Warehouse—Nonvolatile

- A physically separate store of data transformed from the operational environment

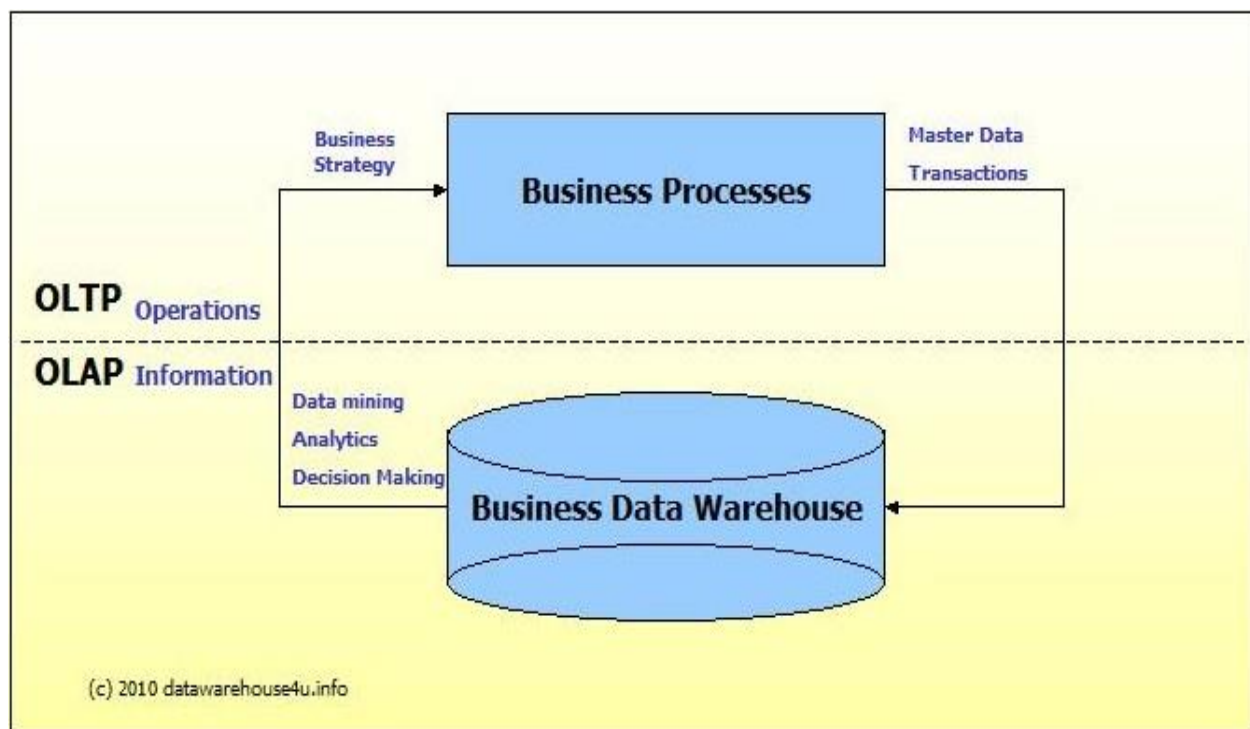
- Operational update of data does not occur in the data warehouse environment
 - Does not require transaction processing, recovery, and concurrency control mechanisms
 - Requires only two operations in data accessing:
 - initial loading of data and access of data

OLTP vs. OLAP

Divide IT systems into transactional (OLTP) and analytical (OLAP).

In general, assume that OLTP systems provide source data to data warehouses, whereas OLAP systems help to analyze it.

- OLTP (on-line transaction processing)
 - Major task of traditional relational DBMS
 - Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.
- OLAP (on-line analytical processing)
 - Major task of data warehouse system
 - Data analysis and decision making



- OLTP (On-line Transaction Processing) is characterized by a large number of short on-line transactions (INSERT, UPDATE, DELETE). The main emphasis for OLTP systems is put on very fast query processing, maintaining data integrity in multi-access environments and an effectiveness measured by number of transactions per second. In OLTP database there is detailed and current data, and schema used to store transactional databases is the entity model (usually 3NF).

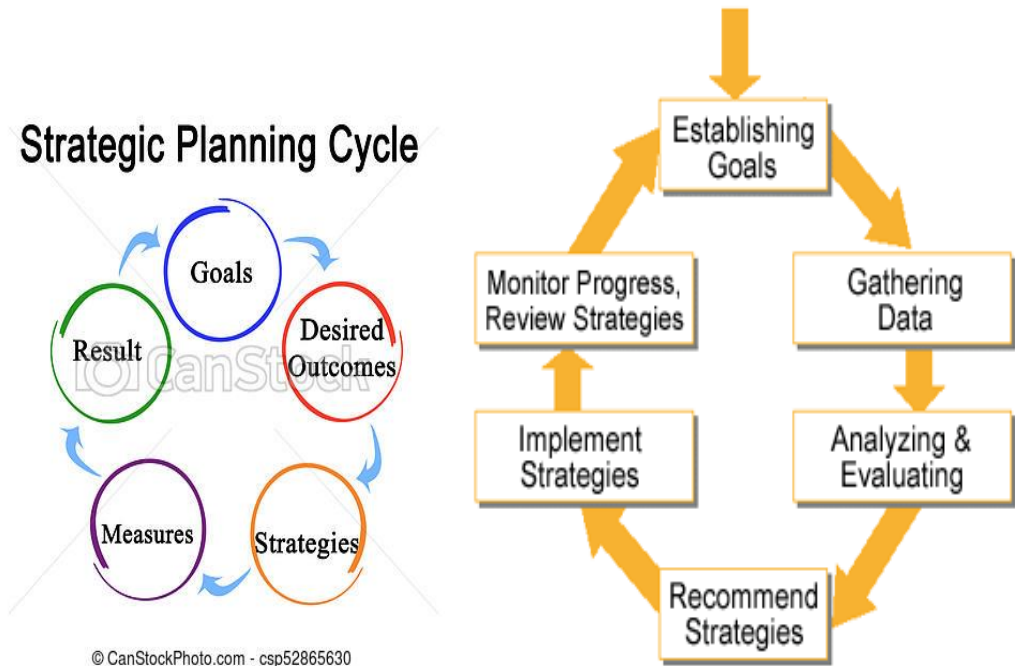
- OLAP (On-line Analytical Processing) is characterized by relatively low volume of transactions. Queries are often very complex and involve aggregations. For OLAP systems a response time is an effectiveness measure. OLAP applications are widely used by Data Mining techniques. In OLAP database there is aggregated, historical data, stored in multi-dimensional schemas (usually star schema).

	OLTP System Online Transaction Processing (Operational System)	OLAP System Online Analytical Processing (Data Warehouse)
Source of data	Operational data; OLTPs are the original source of the data.	Consolidation data; OLAP data comes from the various OLTP Databases
Purpose of data	To control and run fundamental business tasks	To help with planning, problem solving, and decision support
What the data	Reveals a snapshot of ongoing business processes	Multi-dimensional views of various kinds of business activities
Inserts and Updates	Short and fast inserts and updates initiated by end users	Periodic long-running batch jobs refresh the data
Queries	Relatively standardized and simple queries Returning relatively few records	Often complex queries involving aggregations
Processing Speed	Typically very fast	Depends on the amount of data involved; batch data refreshes and complex queries may take many hours; query speed can be improved by creating indexes
Space Requirements	Can be relatively small if historical data is archived	Larger due to the existence of aggregation structures and history data; requires more indexes than OLTP
Database Design	Highly normalized with many tables	Typically de-normalized with fewer tables; use of star and/or snowflake schemas
Backup and Recovery	Backup religiously; operational data is critical to run the business, data loss is likely to entail significant monetary loss and legal liability	Instead of regular backups, some environments may consider simply reloading the OLTP data as a recovery method

CHAPTER 5: PLANNING FOR IS

INFORMATION SYSTEM PLANNING

- Business planning – the process of identifying the firm's goals, objectives, and priorities + developing action plans for accomplishing them.
- Information systems planning – the part of business planning concerned with developing the firm's information systems resources



CHALLENGES IN BUSINESS PLANNING

- Foreseeing and assessing opportunities
- Assuring consistency with organizational plans and objectives
- Building systems
- Maintaining information system performance
- Collaborating with IT professionals y

PRINCIPLES FOR IS PLANNING

- Support the firm's business strategy with appropriate technical architecture
- Evaluate technology as a component of a larger system
- Recognize life cycle costs, not just acquisition costs

WHAT IS INFORMATION SYSTEM PLANNING

- Information system planning is a formal process that develops plan for developing and managing information systems that will support goals of an organization.
 - Information System plan includes:
 - Activities planner believes will help achieve goals.
 - Program for monitoring real-world progress.
 - Means for implementing changes in the plan.

WHY PLAN?

- To obtain resources
 - Financial
 - Facilities – “Capacity planning”
 - Staff
- To align Information System with the business
- To identify needed applications
- To establish goal, schedule, and milestone in order to track progress
- To provide an opportunity for communication with top management and user management

APPROACHES TO PLANNING

- Top-down Planning
 - Focuses on organizational goals first, then on the needs of business units
- Bottom-up Planning
 - Focuses on needs of business units first, then on organizational goals

INFORMATION SYSTEM PLANNING PROCESS

- Establish a mission statement
- Assess the environment
- Set goals and objectives
- Derive strategies and policies
- Develop long-, medium-, and short-range plans implement plans and monitor results

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Establish a Mission Statement

- These are services that you are responsible for; it is your place in the organization.
- It is not what you are supposed to achieve, it is who you are and what you do in the company.

Goals and Objectives

- Set goals – what do you want to achieve?
- Set objectives – what are your specific, measurable targets?

Derive strategies and policies

- Strategies for
 - Technology focus
 - Personnel and career development
 - Aligning with the company
 - Funding criteria; how much to spend on IT?
- Policies for
 - Funding criteria; how much to spend on IT?
 - Allocation criteria; priority setting
 - Organizational arrangements
 - Use of outside IT services, outsourcing
 - Selling IT services to outside organizations

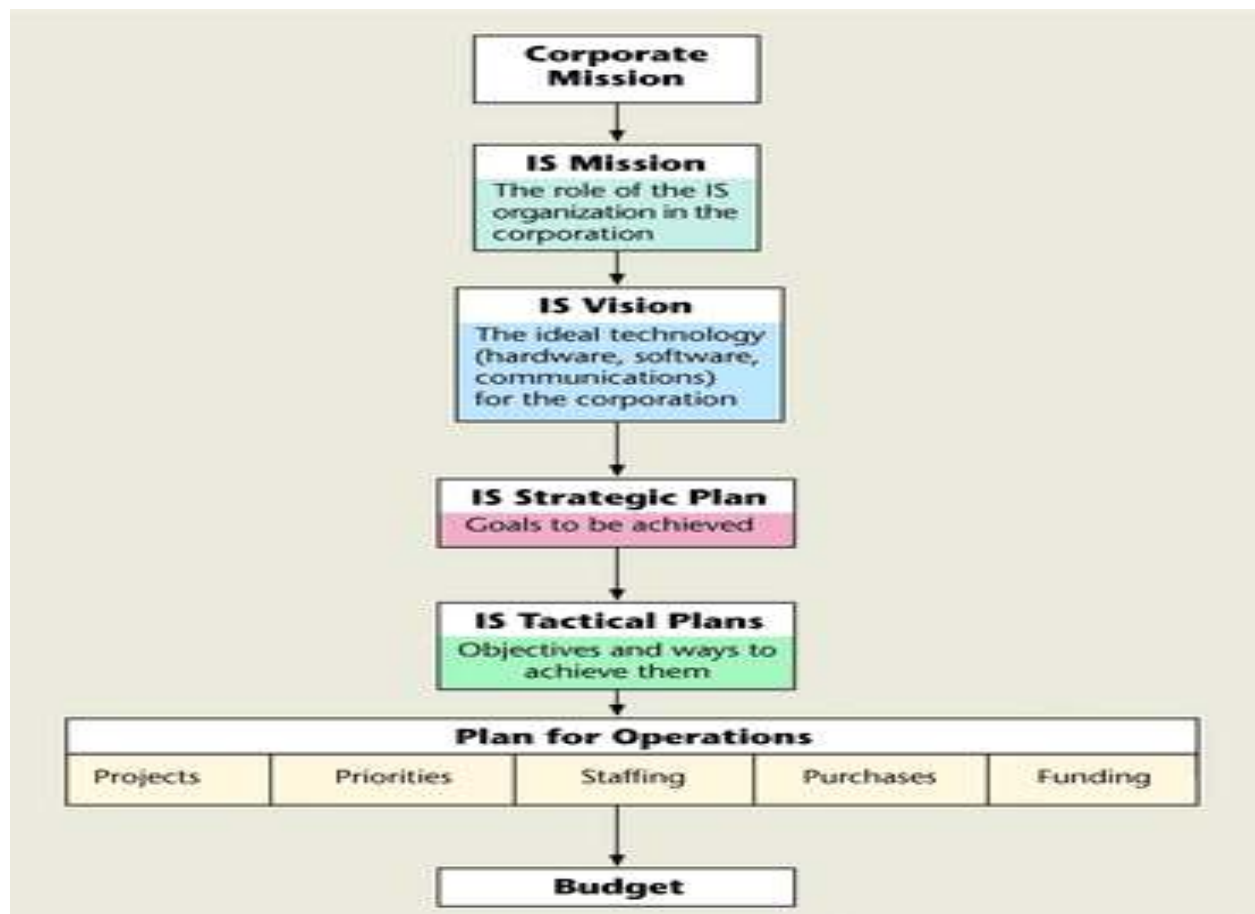
Develop long-, medium-, and short-range plans

- Short-Range – the next year, the next budget period; developing and operating current systems
- Medium-Range – committing to development efforts for applications that will take more than one year to complete; meeting management's current information needs, projected into the future for as many years as needed to complete them. This is what most organizations call "Long-Range Planning."
- Long-Range planning – preparing for management's future information needs. These are not application specific; they are investments in infrastructure ; it is creating an information architecture.

WHAT ARE KEY ELEMENTS OF IS PLANNING

- Key elements of an IS Plan are
 - Corporate mission statement
 - Vision for IT within organization
 - IS strategic and tactical plans
 - Operations plan to achieve mission and vision
 - Budget to ensure resources are available

STEPS OF INFORMATION SYSTEMS PLANNING



INFORMATION SYSTEMS PLANNING

- The Corporate and IS Mission Statements
 - The corporate mission statement details the purpose of the organization and its overall goals.
 - IS mission statement outlines the purpose of ISs in the organization.

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- The IT Vision
 - Wish list of what IS managers would like to see in terms of hardware, software, and communications, to contribute to goals of the organization
- **STRATEGIC IS PLANNING**
 - IS Strategic plan details what is to be achieved
 - Strategic plans are designed with the entire organization in mind and begin with an organization's mission.
 - Essentially, strategic plans look ahead to where the organization wants to be in three, five, even ten years. Strategic plans, provided by top-level managers, serve as the framework for lower-level planning.
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STRATEGIC INFORMATION SYSTEM

- Strategic Information System is a system that helps companies alter their business strategy. It is used to accelerate the reaction time to environmental changes and aid the company in achieving a competitive advantage over its competitors.
- They help in producing low cost quality products.

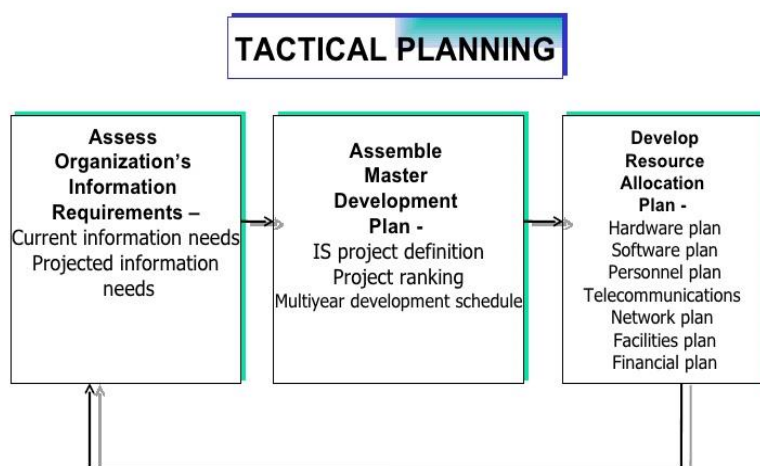
- The strategic role of IS involves using IT to develop products, services, and capabilities that give company major advantages over the competitive forces it faces in the global marketplace.

TYPICAL STRATEGIC-LEVEL JOB TITLES

- Top-level Managers
- CEOs or Presidents
- General Manager
- Corporate Boards
- Steering Committee
- Board of Directors
 - will design and execute strategic plans to paint a picture of the desired future and long-term goals of the organization.

TACTICAL IS PLANNING

- IS Tactical plan describes how goals will be met and by when
- Tactical IS Plan performed by middle managers responsible for acquisition and allocation of resources for projects according to tactical plans, set out for one or two years.
- Tactical IS Planning is evaluates current and projected information needs of the organization, prioritizes IS development projects, and develops allocation plans for financial and technology resources.



What are important factors in IS tactical planning

- Important Factors in IS Tactical Planning
 - Flexibility
 - Compatibility
 - Connectivity
 - Scalability
 - Standardization
 - Total Cost of Ownership

Typical Tactical-Level Job Titles

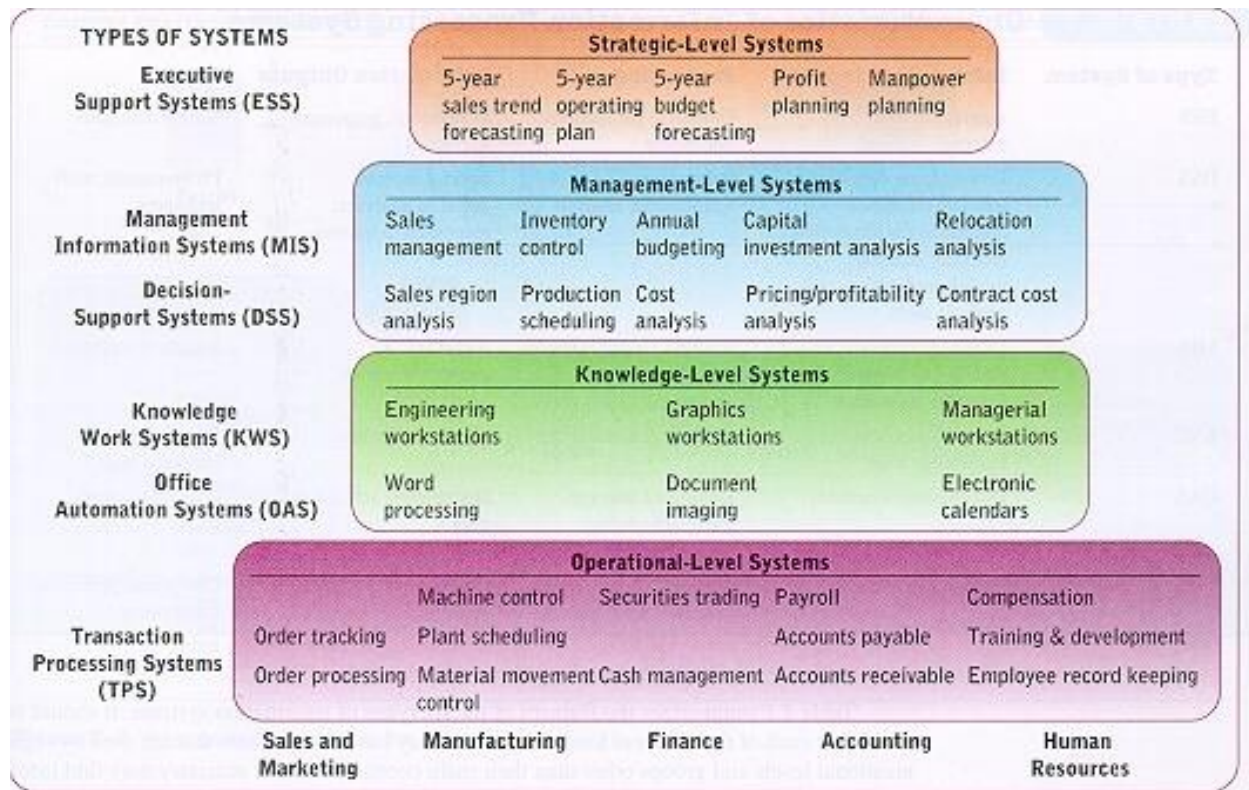
- Advertising manager
- Personnel manager
- Creative director
- Manager of information systems
- Communications director
- Chief Financial Officer

OPERATIONAL IS PLANNING

- Operational Plan prepared by a component of an organization that clearly defines actions it will take to support the strategic objectives and plans of upper management.
- Operational IS Planning develops plans such as annual operating budgets and individual IS project plans.
- Operations IS Planning performed by supervisors of smaller work units concerned with planning and control of short-term (typically, a week or six months) budgets and schedules.



Hierarchy of Planning



Three levels of planning

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Type of Plan	Created By	Scope	Includes	Level of Detail
Strategic Plan	Top Management	Entire organization	Mission of the company, future goals and ambitions	Very broad and general
Tactical Plan	Mid-level Management	Single area of the business as a whole (e.g. a division of the company)	Specific actions to support or work towards the Strategic Plan	Specific actions and ideas, but not very detailed
Operational Plan	Low-level Management	A unit within a single area of the business (e.g. a department within a division)	Specific plans for low level and day-to-day activities and processes that will support and enable the Tactical Plan	Extremely detailed (who, what, where and when)

CHAPTER 6: CHANGE MANAGEMENT AND CRITICAL SUCCESS FACTOR

ORGANIZATION TRANSFORMATION

- Major organizational changes such as transformation to e-business

are referred to as organization transformation.

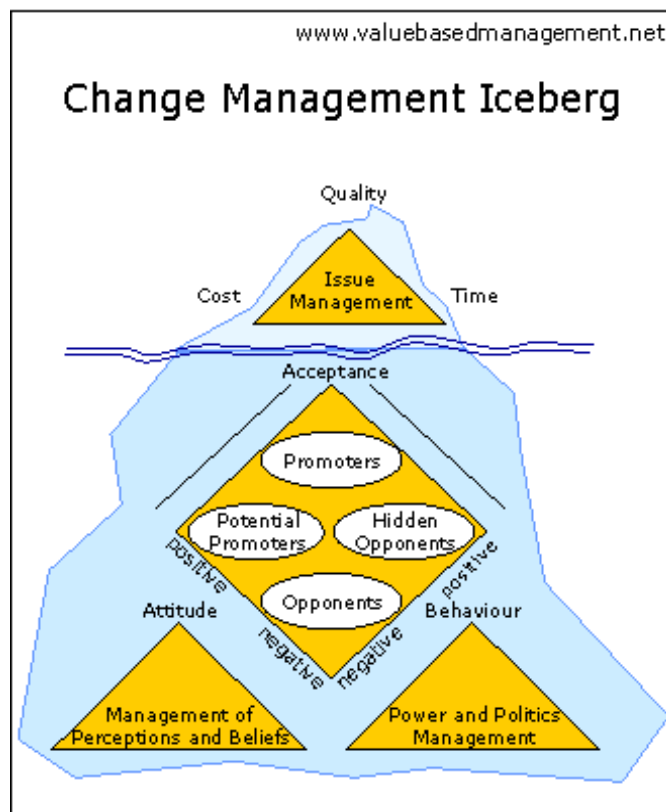
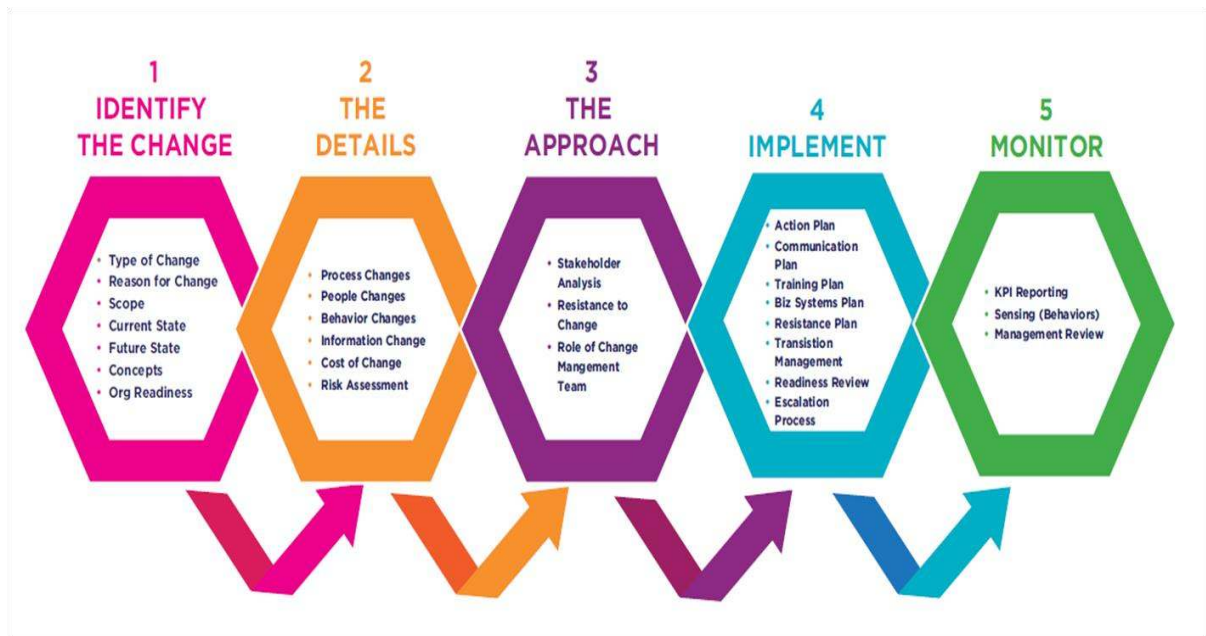
This process usually requires change management

- refers to an organization with a “new face,” whose business processes, structure, strategy, and procedures are completely different from the old one. Such a radical transformation can be a lengthy, expensive, and complex process, which may involve organizational learning, changes in management and personnel, creation of a new structure, and employee retraining.

CHANGE MANAGEMENT

- Refers to the implementation, control and guidelines to introduce change into organizations. Changing business processes, organizational structure and operating procedures are interrelated and depending upon the magnitude of the change can be met with employee resistance. Since change is a learning process its impact can be minimized if properly managed.





The CM Iceberg is a representation of some of the key forces that are in play in an organization facing change.

Project management typically focuses on the top portion of the iceberg...Cost, Time, Quality/Scope.

CM is focused on the focused on the less visible elements below the surface.

In general, people fall into 4 major categories when it comes to change. These categories are represented by the 4 ovals...

At the bottom of the iceberg, you have to very dynamic and powerful forces. The first is Perceptions and Beliefs...it really doesn't matter what the reality of the situation is what's important is what people BELIEVE because that's their reality. The second is Power and Politics.....doors can either be easily opened or soundly shut depending on how this force is managed.

WHY IS CHANGE MANAGEMENT IMPORTANT?

Possible Responses:

- Productivity declines as people become more consumed with the change being introduced.
- Passive resistance festers.
- Active resistance emerges and sabotages the change.
- Valued employees leave the organization, a very costly proposition in terms of the lost contribution and the cost to replace them.
- Employees become disinterested in the current state and the future state.
- Employees begin arguing about change and the direction of the company.
- People are left to wonder why the change is happening.
- More people begin taking sick days or not showing up for work.
- People find work-arounds to avoid implementing the new way of doing things.
- Employees revert back to the old way of doing things.
- Changes are not fully implemented.
- Changes are scrapped and cancelled due to the lack of support throughout the organization
- Divides are created in the organization between 'us' and 'them'.
- The organization builds a history of failed and painful changes.
- Many types of risk are created - risk to the project, to the organization, to the employees involved and to the individuals supporting or chartering the change.

BENEFITS OF EFFECTIVELY MANAGED CHANGE

- Employees have a solid understanding of why change is happening.
- Employees engage in both the solution and the change.
- Training is used to build knowledge after employees have made the personal decision to support the change.
- Resistance is identified and dealt with early in the process.
- Senior leaders demonstrate their own and the organization's commitment to the change.

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- Communications are segmented and customized for different audiences, answering the questions that they care about.
- Momentum is built throughout different areas and levels within the organization.
- Changes are less painful to the organization and to the employees.
- A coalition of support among senior leaders and managers creates momentum throughout the organization.
- Probability of meeting project objectives is increased.
- The organization begins to build a history of successful change, creating a better 'backdrop' for the next change initiative.

CHANGE MANAGEMENT TOOLS

CM Tools:

- Project Charter
- Communication Plan
 - Multiple Communication Channels (e.g. Newsletters, SharePoint, Project Webpage, meetings, social media, UP/DOWN etc....'Living under a rock')
- Business Process Documentation
- Business Process Documentation
 - This was the first project that we used BPD...
 - Help you identify areas with significant change. May need more handholding...
 - identify pain points, bottlenecks, WASTE.
- Our Leadership feels so strongly about the benefits of BPD, that no new large technology projects will be considered unless the processes have been documented.
- Training Plan

GOALS OF CHANGE MANAGEMENT

- Convergence of User and System Readiness
- Optimize Organization's Social (Cultural) and Technical Infrastructures
- Create a Learning Organization
 - Continuous Improvement
 - Customer Focus
 - Adaptable to changes in the environment (external and internal)

KEY PRINCIPLES OF CHANGE MANAGEMENT

- Steering Committee Must Have Product and Process Champions or Stakeholders
- Communication needs to multi-channeled (up is as valued as down)

- Conflict resolution and knowledge acquisition are collaborative (information sharing not hoarding)
- Implementation Team (change management team) has a structure and roles of individuals are defined
- Resistance to Change is always there but may be hard to detect until after Go Live (Be Proactive)

CHALLENGES AND PITFALLS

- Change process must be planned and managed
- Magnitude of change is ALWAYS greater than imagined
- Resistance to change appears in different forms -- must be addressed
- In global organizations cultural factors need special consideration
- Change in organizations occurs in stages -- need strategies for EACH stage

STAGES OF CHANGE

- **Boston consulting Group 5 Stage Model:**
 - Stagnation (building desire to change)
 - Anticipation (acceptance of change is highest here)
 - Implementation (critical time for commitment -- most pain occurs here)
 - Determination (expectations get real -- people begin to incorporate change into daily routine)
 - Fruition (acceptance begins to climb back up)
- **Lewins' 3 Stage Model: Unfreeze, Change and Re-freeze**
- Unfreezing:
 - Develop awareness of change
 - Users see or have a need to change (internal felt need)
 - Driving forces for change outweigh the Restraining Forces
- Change:
 - Involve users in project team (as is --> to be processes)
 - Train the users

- Develop in-house gurus -- change agents that can help resisters

. Refreeze

- the changes are used all the time, and that they are incorporated into everyday business.

MANAGERIAL ISSUES

- Getting IT ready for the future. Planning—is one of the most challenging and difficult tasks facing all of management, including IS management. Each of the four steps of the IT strategic planning process—strategic planning, information requirements analysis, resource allocation, and project planning—presents its own unique problems. Yet, without planning, or with poor planning, the organization may be doomed.
- Organizing for planning. Many issues are involved in planning: What should be the role of the ISD? How should IT be organized? Staffed? Funded? How should human resources issues, such as training, benefits, and career paths for IS personnel, be handled? What about the environment? The competition? The economy? Governmental regulations? Emerging technologies? What is the strategic direction of the host organization? What are its key objectives? Are they agreed upon and clearly stated? Finally, with these strategies and objectives and the larger environment, what strategies and objectives should IS pursue? What policies should it establish? What type of information architecture should the organization have: centralized or not centralized? How should investments in IT be justified? The answer to each of these questions must be tailored to the particular circumstances of the ISD and the larger organization of which it is a part.
- Fitting the IT architecture to the organization. Management of an organization may become concerned that its IT architecture is not suited to the needs of the organization. In such a case, there has likely been a failure on the part of the IT technicians to determine properly the requirements of the organization. Perhaps there has also been a failure on the part of management to understand the type and manner of IT architecture that they have allowed to develop or that they need.
- IT architecture planning. IT specialists versed in the technology of IT must meet with business users and jointly determine the present and future needs for the IT architecture. In some cases, IT should lead (e.g., when business users do not understand the technical implications of a new technology). In other cases, users should lead (e.g., when technology is to be applied to a new business opportunity). Plans should be written and published as part of the organizational strategic plan and as part of the IT strategic plan. Plans should also deal with training, career implications, and other secondary infrastructure issues.
- IT policy. IT architectures should be based on corporate guidelines or principles laid out in policies. These policies should include the roles and responsibilities of IT personnel and users, security issues, cost-benefit analyses for evaluating IT, and IT architectural

goals. Policies should be communicated to all personnel who are managing or directly affected by IT.

- Ethical and legal issues. Conducting interviews for finding managers' needs and requirements must be done with full cooperation. Measures to protect privacy must be taken. In designing systems one should consider the people in the system. Reengineering IT means that some employees will have to completely reengineer themselves. Conducting a supply chain or business process reorganization may result in the need to lay off, retrain, or transfer employees. Should management notify the employees in advance regarding such possibilities? And what about those older employees who are difficult to retrain?
- Other ethical issues may involve sharing of computing resources (in a client/server environment, for example) or of personal information, which may be part of the new organizational culture. Finally, individuals may have to share computer programs that they designed for their departmental use, and may resist doing so because they consider such programs their intellectual property. Appropriate planning must take these and other issues into consideration.
- When using business intelligence to find out what competitors are doing, companies may be engaged in unethical tactics such as pressuring competitors' employees to reveal information, or using software that is the intellectual property of other companies (frequently without the knowledge of these other companies).
- IT strategy. In planning IT it is necessary to examine three basic strategies:
 - Be a leader in technology. However, there is a high development cost of new technologies and high probability of failures.
 - Be a follower. This is a risky strategy because you may be left behind.
 - Be an experimenter, on a small scale. This way you minimize your research and development investment and the cost of failure.
- Integration: The role of IT in redesign and BPR. Almost all major supply chain management (SCM) and/or BPR projects use IT. However, it is important to remember that in most cases the technology plays a supportive role. The primary role is organizational and managerial in nature. On the other hand, without IT, most SCM and BPR efforts do not succeed.
- Failures. One of the lessons from the history of IT is that very big projects have a tendency to fail when expectations exceed real capabilities. One of the reasons for failure is a miscalculation of the required amount of IT. It simply may be too expensive to rebuild and retool the IT infrastructure and adjust applications that are necessary for BPR. The solution may be instead to defer the BPR and use incremental improvements, or to reengineer only the most critical processes.

CRITICAL SUCCESS FACTORS (CSFS)

BY: ER. ANKU JAISWAL

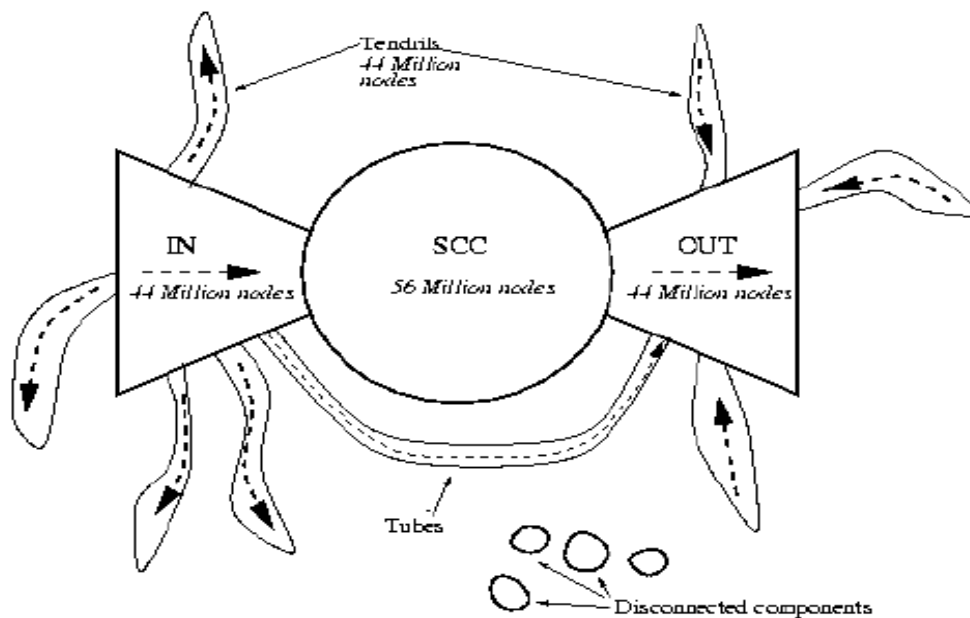
- CSFs are those few things that must go right in order to ensure the organization's survival and success. Critical success factors vary by industry categories—manufacturing, service, or government—and by specific industries within these categories. Sample questions asked in the CSF approach are:
 - What objectives are central to your organization?
 - What are the critical factors that are essential to meeting these objectives?
 - What decisions or actions are key to these critical factors?
 - What variables underlie these decisions, and how are they measured?
 - What information systems can supply these measures?

CHAPTER 7: WEB BASED INFORMATION SYSTEM AND NAVIGATION

Structure of the Web

- The growth of www.
- Have no engineered architecture or structure, but rather a virtual network of content and hyperlinks.
- a large, strongly connected core in which every page can reach every other by a path of hyperlinks.

Different nodes are upstream, downstream and tendrils nodes.



Different nodes:

- In the SCC
- In the “inbound” part
- In the “outbound” part
- Tendrils
- Disconnected nodes

In 1999, after the Web had been growing for the better part of a decade, Andrei Broder and his colleagues set out to build a global map of the Web, using strongly connected components.

The Bow-Tie Structure. The second step is to position all the remaining SCCs in relation to the giant one. This involves classifying nodes by their ability to reach and be reached from the giant SCC. The first two sets in this classification are the following.

(1) IN: nodes that can reach the giant SCC but cannot be reached from it — i.e., nodes that are “upstream” of it.

(2) OUT: nodes that can be reached from the giant SCC but cannot reach it — i.e., nodes are “downstream” of it.

3. The strongly connected components or disconnected components of an arbitrary directed graph form a partition into subgraphs that are themselves strongly connected.

4. Tendrils: The “tendrils” of the bow-tie consist of (a) the nodes reachable from IN that cannot reach the giant SCC, and (b) the nodes that can reach OUT but cannot be reached from the giant SCC.

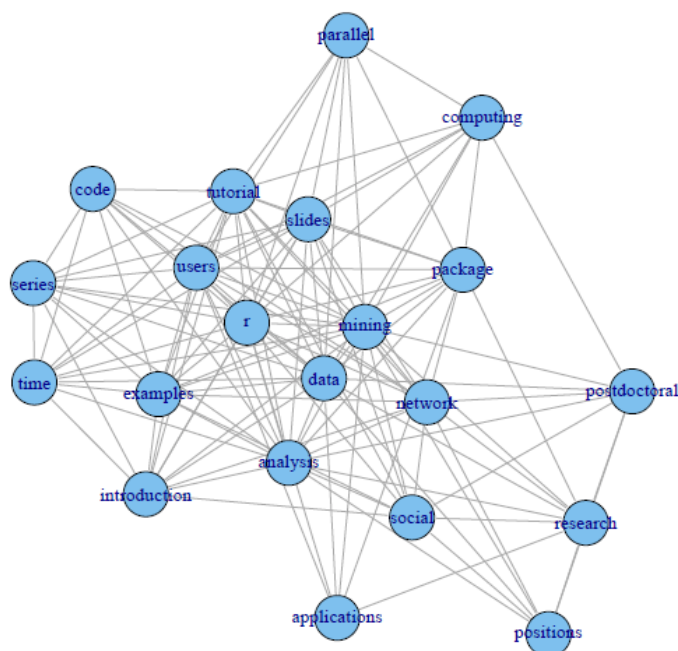
5 Disconnected: Finally, there are nodes that would not have a path to the giant SCC even if we completely ignored the directions of the edges. These belong to none of the preceding categories.

LINK ANALYSIS

Link analysis is a data analysis technique used in network theory that is used to evaluate the relationships or connections between network nodes. These relationships can be between various types of objects (nodes), including people, organizations and even transactions.

Link analysis is essentially a kind of knowledge discovery that can be used to visualize data to allow for better analysis, especially in the context of links, whether Web links or relationship links between people or between different entities. Link analysis is often used in search engine optimization as well as in intelligence, in security analysis and in market and medical research.

- a means of finding authoritative, relevant sources on the Web has proven useful in the design of improved search engines
- a data analysis technique used to evaluate relationships (connections) between nodes
- Leading to improved methods for accessing and understanding the available information
- Web pages can be defined as hubs and authorities.
- Characteristic patterns for to identify communities of pages on the same topic



SEARCHING THE WEB

- Searching in the web is done by search engine
- A search engine operates in the following order:
- Web crawling
- Indexing
- Searching
- Web search engines work by storing information about many web pages, which they retrieve from the HTML markup of the pages. These pages are retrieved by a Web crawler (sometimes also known as a spider) — an automated Web crawler which follows every link on the site. The site owner can exclude specific pages by using robots.txt.
- The search engine then analyzes the contents of each page to determine how it should be indexed (for example, words can be extracted from the titles, page content, headings, or special fields called meta tags). Data about web pages are stored in an index database for use in later queries. A query from a user can be a single word. The index helps find information relating to the query as quickly as possible. Some search engines, such as Google, store all or part of the source page (referred to as a cache) as well as information about the web pages, whereas others, such as AltaVista, store every word of every page they find. This cached page always holds the actual search text since it is the one that was actually indexed, so it can be very useful when the content of the current page has been updated and the search terms are no longer in it. This problem might be considered a mild form of linkrot, and Google's handling of it increases usability by satisfying user expectations that the search terms will be on the returned webpage. This satisfies the

principle of least astonishment, since the user normally expects that the search terms will be on the returned pages. Increased search relevance makes these cached pages very useful as they may contain data that may no longer be available elsewhere.

- High-level architecture of a standard Web crawler
- When a user enters a query into a search engine (typically by using keywords), the engine examines its index and provides a listing of best-matching web pages according to its criteria, usually with a short summary containing the document's title and sometimes parts of the text. The index is built from the information stored with the data and the method by which the information is indexed. From 2007 the Google.com search engine has allowed one to search by date by clicking "Show search tools" in the leftmost column of the initial search results page, and then selecting the desired date range. Most search engines support the use of the boolean operators AND, OR and NOT to further specify the search query. Boolean operators are for literal searches that allow the user to refine and extend the terms of the search. The engine looks for the words or phrases exactly as entered. Some search engines provide an advanced feature called proximity search, which allows users to define the distance between keywords. There is also concept-based searching where the research involves using statistical analysis on pages containing the words or phrases you search for. As well, natural language queries allow the user to type a question in the same form one would ask it to a human. A site like this would be ask.com.
- The usefulness of a search engine depends on the relevance of the result set it gives back. While there may be millions of web pages that include a particular word or phrase, some pages may be more relevant, popular, or authoritative than others. Most search engines employ methods to rank the results to provide the "best" results first. How a search engine decides which pages are the best matches, and what order the results should be shown in, varies widely from one engine to another. The methods also change over time as Internet usage changes and new techniques evolve. There are two main types of search engine that have evolved: one is a system of predefined and hierarchically ordered keywords that humans have programmed extensively. The other is a system that generates an "inverted index" by analyzing texts it locates. This first form relies much more heavily on the computer itself to do the bulk of the work.
- Most Web search engines are commercial ventures supported by advertising revenue and thus some of them allow advertisers to have their listings ranked higher in search results for a fee. Search engines that do not accept money for their search results make money by running search related ads alongside the regular search engine results. The search engines make money every time someone clicks on one of these ads.

WEB NAVIGATION

- Web navigation refers to the process of navigating a network of information resources in the World Wide Web, which is organized as a hypertext or hypermedia. The user interface that is used to do so is called a web browser. A central theme in web design is the development of a web navigation interface that maximizes usability. A website's overall navigational scheme includes several navigational pieces such as global, local,

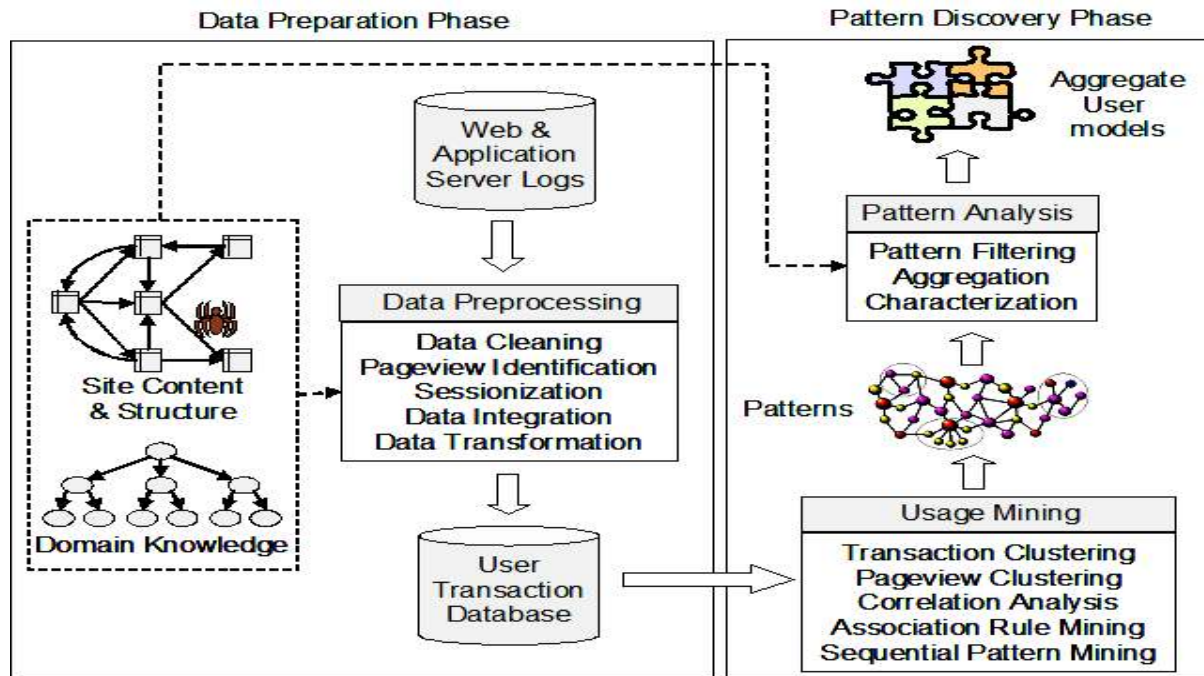
supplemental, and contextual navigation; all of these are vital aspects of the broad topic of web navigation. Hierarchical navigation systems are vital as well since it is the primary navigation system. It allows for the user to navigate within the site using levels alone, which is often seen as restricting and requires additional navigation systems to better structure the website.^[4] The global navigation of a website, as another segment of web navigation, serves as the outline and template in order to achieve an easy maneuver for the users accessing the site, while local navigation is often used to help the users within a specific section of the site. All these navigational pieces fall under the categories of various types of web navigation, allowing for further development and for more efficient experiences upon visiting a webpage.

WEB MINING

- Web mining is the application of data mining techniques to discover patterns from the Web.
- Web mining can be divided into three different types, which are
 - Web usage mining,
 - Web content mining
 - Web structure mining.

WEB USAGE MINING

- Web usage mining is the process of extracting useful information from server logs e.g. use Web usage mining is the process of finding out what users are looking for on the Internet.
- Some users might be looking at only textual data, whereas some others might be interested in multimedia data.
- Web Usage Mining is the application of data mining techniques to discover interesting usage patterns from Web data in order to understand and better serve the needs of Web-based applications.
- Usage data captures the identity or origin of Web users along with their browsing behavior at a Web site.



Classification of Web Usage mining

- Web Server Data:

The user logs are collected by the Web server. Typical data includes IP address, page reference and access time.

- Application Server Data:

Commercial application servers have significant features to enable e-commerce applications to be built on top of them with little effort. A key feature is the ability to track various kinds of business events and log them in application server logs.

- Application Level Data:

New kinds of events can be defined in an application, and logging can be turned on for them thus generating histories of these specially defined events. It must be noted, however, that many end applications require a combination of one or more of the techniques applied in the categories above.

Pros & Cons of Web Usage mining

- Government agencies are using this technology to classify threats and fight against terrorism.
- The predicting capability of mining applications can benefit society by identifying criminal activities.
- Better customer relationship by giving them exactly what they need.

- attract and retain customers
- Cons
 - Invasion of Privacy

COLLABORATIVE FILTERING

- Collaborative filtering is the process of filtering for information or patterns using techniques involving collaboration among multiple agents, viewpoints, data sources, etc.
- Collaborative filtering is a method of making automatic predictions (filtering) about the interests of a user by collecting preferences or taste information from many users
- Applications of collaborative filtering typically involve very large data sets.
- It is applied to many different kinds of data including:
 - sensing and monitoring data, such as in mineral exploration, environmental sensing over large areas or multiple sensors
 - financial data, such as financial service institutions that integrate many financial sources
 - in electronic commerce and web applications where the focus is on user data, etc
- Collaborative filtering algorithms often require
 - users' active participation
 - an easy way to represent users' interests to the system
 - algorithms that are able to match people with similar interests.

Methodology

- Collaborative filtering systems can be reduced to two steps:
 - Look for users who share the same rating patterns with the active user (the user whom the prediction is for).
 - Use the ratings from those like-minded users found in step 1 to calculate a prediction for the active user
 - A specific application of this is the user-based Nearest Neighbor algorithm.
- Alternatively, item-based collaborative filtering invented by Amazon.com (users who bought x also bought y), proceeds in an item-centric manner:
 - Build an item-item matrix determining relationships between pairs of items

- Infer the tastes of the current user by examining the matrix and matching that user's data

Challenges in Collaborative Filtering

- Data sparsity : based on large datasets ,matrix used for collaborative filtering could be extremely large and sparse, which brings about the challenges in the performances of the recommendation.
- Scalability : Eg:millions of customers then complex
- Synonyms: tendency of a number of the same or very similar items to have different names or entries. Most recommender systems are unable to discover this latent association and thus treat these products differently.
- Grey sheep: refers to the users whose opinions do not consistently agree or disagree with any group of people and thus do not benefit from collaborative filtering.
- Shilling attacks: people may give lots of positive ratings for their own items and negative ratings for their competitors.

COLD START PROBLEM

- Cold start is a potential problem in computer-based information systems which involve a degree of automated data modelling.
- It concerns the issue that the system cannot draw any inferences for users or items about which it has not yet gathered sufficient information.
- In the collaborative filtering approach, the recommender system would identify users who share the same preferences (e.g. rating patterns) with the active user, and propose items which the like-minded users favoured (and the active user has not yet seen). Due to the cold start problem, this approach would fail to consider items which no-one in the community has rated previously.
- Solution
 - In scenarios involving interface agents, the cold start problem may be overcome by introducing an element of collaboration amongst agents assisting various users (Community based recommendation).
 - This way, novel situations may be handled by requesting other agents to share what they have already learnt from their respective users

RECOMMENDER SYSTEMS

- This system is a subclass of information filtering system that seek to predict the 'rating' or 'preference' that user would give to an item.
- most popular used on movies, music, news, books, research articles, search queries, social tags, and products

- Produces recommendation by one of two ways:

1. Through Collaborative filtering

Through Content based filtering

Collaborative filtering:

- build a model from a user's past behavior and use that model to predict items (or ratings for items) that the user may have an interest in.

Content based filtering:

- utilize a series of discrete characteristics of an item in order to recommend additional items with similar properties.

COLLECTIVE INTELLIGENCE

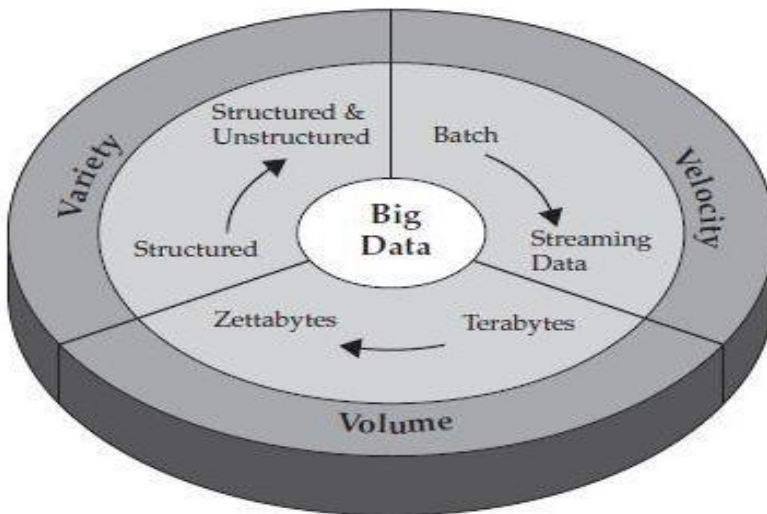
- It is a shared or group intelligence that emerges from the collaboration, collective efforts, and competition of many individuals
- It is used in consensus decision making
- According to Pierre Lévy "It is a form of universally distributed intelligence, constantly enhanced, coordinated in real time, and resulting in the effective mobilization of skills."
- Collective intelligence is a mass collaboration. In order for this concept to happen, four principles need to exist:
- Openness: Sharing ideas and intellectual property: By allowing others to share ideas, it gains significant improvement.
- Peering: Peering where users are free to modify and develop programs, provided that they make it available for others.
- Sharing: Companies have started to share some ideas while maintaining some degree of control over others.
- Acting Globally: The internet is widespread, therefore a company has no geographical boundaries and may access new markets, ideas and technology.

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CHAPTER 8: SCALABLE AND EMERGING INFORMATION SYSTEM TECHNIQUES

BIG DATA

- Big Data applies to information that can't be processed or analyzed using traditional processes or tools.



IBM characterizes Big Data by its volume, velocity, and variety—or simply,

WHAT IS SCALABILITY IN TERMS OF IS?

- The ability of a system, network, or process to handle a growing amount of work in a capable manner or the ability to accommodate growth
- There are 2 types of Scaling:
 - Vertical Scaling; where resources are added to a single node
 - Horizontal Scaling; where new nodes are added to system

TECHNIQUES FOR SCALABILITY IN IS

- Distributed Information System
- Grid Information System
- Parallel Information Processing System
- P2P Systems
- Web Information Searching and Retrieval
- Content Delivery Networks (CDN)
- Mobile and Wireless Database System

TECHNIQUES FOR VOLUMINOUS DATA

- Cloud Computing is an efficient method to balance between dealing with voluminous data and keeping costs competitive, is designed to deliver IT services consumable on demand, is scalable as per user need and uses a pay-per-use model.
- Business houses are progressively turn towards retaining core competencies, and shedding the non-core competencies for on-demand technology, business innovation and savings.

CLOUD COMPUTING

- Cloud computing consists of hardware and software resources made available on the internet as managed third-party services.
- These services typically provide access to advanced software applications and high-end networks of server computers.
- Cloud computing is comparable to grid computing, a type of computing where unused processing cycles of all computers in a network are connect to solve problems too intensive for any stand-alone machine.

Cloud Computing: Public Clouds

- Public clouds are made available to the general public by a service provider who hosts the cloud infrastructure.
- A public cloud is the obvious choice when:
 - Your standardized workload for applications is used by lots of people, such as e-mail.
 - You need to test and develop application code.
 - You need incremental capacity (the ability to add compute resources for peak times).
 - You're doing collaboration projects

Cloud Computing: Private Clouds

- Private cloud is cloud infrastructure dedicated to a particular organization
- When is a Private Cloud for you?
 - You need data sovereignty but want cloud efficiencies
 - You want consistency across services
 - You have more server capacity than your organization can use
 - Your data center must become more efficient
 - You want to provide private cloud services

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Cloud Computing: Hybrid Clouds

- Hybrid Clouds are a composition of two or more clouds (private, community or public) that remain unique entities
- Here are a couple of situations where a hybrid environment is best:
 - Your company wants to use a SaaS application but is concerned about security.
 - Your company offers services that are tailored for different vertical markets. You can use a public cloud to interact with the clients but keep their data secured within a private cloud.
 - You can provide public cloud to your customers while using a private cloud for internal IT.

Cloud Computing: Community Clouds

- A community cloud is a multi-tenant cloud service model that is shared among several organizations and that is governed, managed and secured
- Here are a couple of situations where a community cloud environment is best:
 - Government organizations within a state that need to share resources
 - A private HIPAA compliant cloud for a group of hospitals or clinics
 - Telco community cloud for telco DR to meet specific FCC regulations

Types of Cloud Computing

SaaS (Software as a Service): This is the idea of providing a given application to multiple tenants, typically using the browser which supports business applications of host and delivery type as a service. End Customers for instance Google Doc, Myspace.com

Common features of SaaS:

- a) User applications run on cloud infrastructure
- b) Accessible by users through web browser
- c) Suitable for CRM (Customer Resource Management) applications
- d) Supports multi-tenant environment

IaaS (Infrastructure as a Service): These are virtual storage and server options that organizations can access on demand, even allowing the creation of a virtual data center. Delivers computing hardware like Servers, Network, Storage, etc. For instance Rackspace.com, GoGrid.com etc.

Typical features are:

- a)Users use resources but have no control of underlying cloud infrastructure
- b)Users pay for what they use
- c)Flexible scalable infrastructure without extensive pre-planning

Benefit of Cloud Computing

- Reduced Cost : Cloud technology is paid incrementally, saving organizations money.
- Increased Storage: Organizations can store more data than on private computer systems.
- Highly Automated: No longer do IT personnel need to worry about keeping software up to date.
- Flexibility: Cloud computing offers much more flexibility than past computing methods.
- More Mobility: Employees can access information wherever they are, rather than having to remain at their desks.
- Allows IT to Shift Focus: No longer having to worry about constant server updates and other computing issues, government organizations will be free to concentrate on innovation.

MAPREDUCE

- It is programming paradigm used in Hadoop
- It allows massive scalability across large number of servers
- MapReduce actually refers to two separate and distinct tasks that Hadoop programs perform viz. Map and Reduce
- Map Job: takes a set of data and converts it into another set of data
- Reduce Job: takes the output from map as input and combines those tuples into smaller set of tuples.
- MapReduce is a software framework that allows developers to write programs that process massive amounts of unstructured data in parallel across a distributed cluster of processors or stand-alone computers.
- It was developed at Google for indexing Web pages and replaced their original indexing algorithms and heuristics in 2004.

MapReduce and Hadoop Systems

- MapReduce is the heart of Hadoop.
- MapReduce allows data to be distributed across a large cluster, and can distribute out tasks across the data set to work on pieces of it independently, and in parallel.

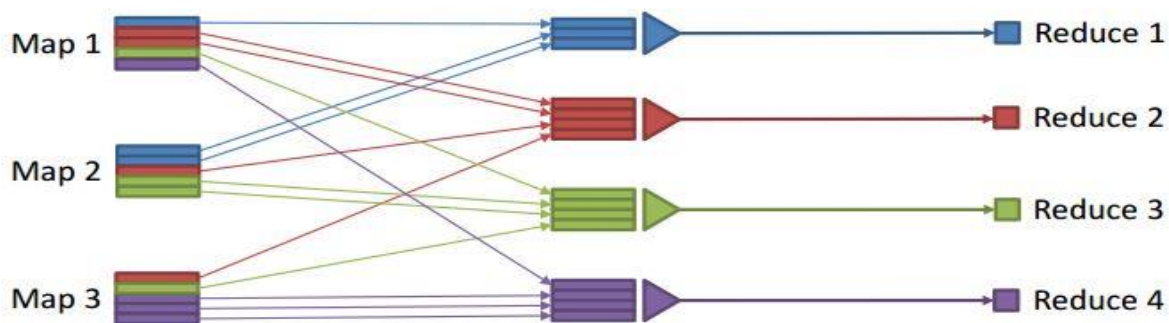
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- This allows big data to be processed in relatively little time.
- Apache has produced an open source MapReduce platform called Hadoop.

Framework is divided into two parts

- Map, a function that parcels out work to different nodes in the distributed cluster.
- Reduce, another function that collates the work and resolves the results into a single value.
- The MapReduce framework is fault-tolerant because each node in the cluster is expected to report back periodically with completed work and status updates. If a node remains silent for longer than the expected interval, a master node makes note and re-assigns the work to other nodes.

MapReduce Interaction



Map functions create a user-defined “index” from source data

- Reduce functions compute grouped aggregates based on index
- Flexible framework
 - users can cast raw original data in any model that they need
 - wide range of tasks can be expressed in this simple framework

HADOOP SYSTEMS

- It is an open-source software framework that supports data-intensive distributed applications
- It supports running applications on large clusters with both reliability and data motion to applications
- Hadoop utilizes Hadoop Distributed File System written in Java.
- Some Prominent Users of Hadoop Systems:
 - Yahoo!

- Facebook
- Amazon
- Google

Twitter

- Developed by Apache as an open source distributed MapReduce platform, based off of Google's MapReduce.
- Runs on a Java architecture framework that supports the processing of large data sets in a distributed computing environment.
- Hadoop allows businesses to process large amounts of data quickly by distributing the work across several nodes.
- Good for Big data sets and on large cluster.

Hadoop System is used by Large Content-Distribution Companies, such as...

Yahoo

- Hadoop is used for many of their tasks, and over 25,000 computers are running Hadoop.

Amazon

- Hadoop is good for Amazon, they have lots of product data, as well as user-generated content to index, and make searchable.

New York Times

Hadoop is used to perform large-scale image conversions of public domain articles

Used by Non-Content-Distribution Companies, such as

- Facebook
- eHarmony

Other early adopters include anyone with big data:

- medical records
- tax records
- network traffic
- large quantities of data

Wherever there is a lot of data, a Hadoop cluster can generally process it relatively quickly.

DATA MANAGEMENT IN THE CLOUD

BY: ER. ANKU JAISWAL

- Data management applications are potential candidates for deployment in the cloud
 - Industry: enterprise database system have significant up-front cost that includes both hardware and software costs
 - Academia: manage, process and share mass-produced data in the cloud
- Many “Cloud Killer Apps” are in fact data-intensive
 - Batch Processing as with MapReduce
 - Online Transaction Processing (OLTP) as in automated business applications
 - Offline Analytical Processing (OLAP) as in data mining or machine learning
 - A database system must implement for it to run well in the cloud, in potential database applications to consider for cloud deployment.
 - Data management applications are best suited for deployment on top of cloud computing infrastructure.
 - Data management is the proper management of a data resource for an organization. Data management consists of a set of theories, concepts, principles, and techniques for properly managing data.
 - The primary objective is to support the business information as needs of the organization.

There are three characteristics of a cloud computing environment.

- Compute power is elastic, but only if workload is parallelizable.
- Data is stored at untrusted host.
- Data is replicated, often across large geographic distances
- Data management is carried out by Cloud Data Management Interface (CDMI)
- CDMI is standard that specifies protocols for self-provisioning, administering and accessing cloud storage

INFORMATION RETRIEVAL

- Information retrieval is the activity of obtaining information resources relevant to an information need from a collection of information resources.
- Searches can be based on metadata or on full-text indexing.

- Automated information retrieval systems are used to reduce what has been called “information overload”.

Information Retrieval in the Cloud

- IR user seeks actively information, pulling at it, by means of querying or browsing.
- In tag querying, user enters one or more tags in the search box to obtain an ordered list of resources which were in relation with these tags.
- When a user is scanning this list, the system also provide a list of related tags (i.e. tags with a high degree of co-occurrence with the original tag), allowing hypertext Browsing.
 - Typically it refers to the automatic (rather than manual) retrieval of documents
 - Information Retrieval System (IRS)
- Information Retrieval is a research-driven theoretical and experimental discipline
 - The focus is on different aspects of the information-seeking process, depending on the researcher’s background or interest:
 - Computer scientist – fast and accurate search engine
 - Librarian – organization and indexing of information
 - Cognitive scientist – the process in the searcher’s mind
 - Philosopher – Is this really relevant ?

Stopwords / Stoplist

- Function words do not bear useful information for IR
 - of, in, about, with, I, although, ...
- Stoplist: contain Stopwords, not to be used as index
 - Prepositions
 - Articles
 - Pronouns
 - Some adverbs and adjectives
 - Some frequent words (e.g. document)
- The removal of stopwords usually improves IR effectiveness
- A few “standard” stoplists are commonly used.

Stemming

- Reason:
 - Different word forms may bear similar meaning (e.g. search, searching): create a “standard” representation for them
- Stemming:
 - Removing some endings of word
 - computer
 - compute
 - computes
 - computing
 - computed
 - computation

LINK ANALYSIS IN CLOUD SETUP

- The web is not just a collection of documents – its hyperlinks are important!
- A link from page A to page B may indicate:
 - A is related to B, or
 - A is recommending, citing, voting for or endorsing B
- Links are either
 - referential – click here and get back home, or
 - Informational – click here to get more detail
- Links effect the ranking of web pages and thus have commercial value.

Some presentation slide from our Students 4th year 2074 batch

1. Balanced Scorecard

BALANCED SCORECARD AND ITS IMPORTANCE

Diwakar Dhungana
Student

The Balanced Scorecard is a management tool that provides stakeholders with a comprehensive measure of how the organization is progressing towards the achievement of its strategic goals.

Balanced scorecard concept

- First published in 1992 by Kaplan and Norton
- The approach is to provide 'balance' to the financial perspective

Balanced scorecard:

- Balances financial and non-financial measures.
- Balances short and long-term measures.
- Should contain just enough data to give a complete picture of organizational performance.
- Leads to strategic focus and organizational alignment.

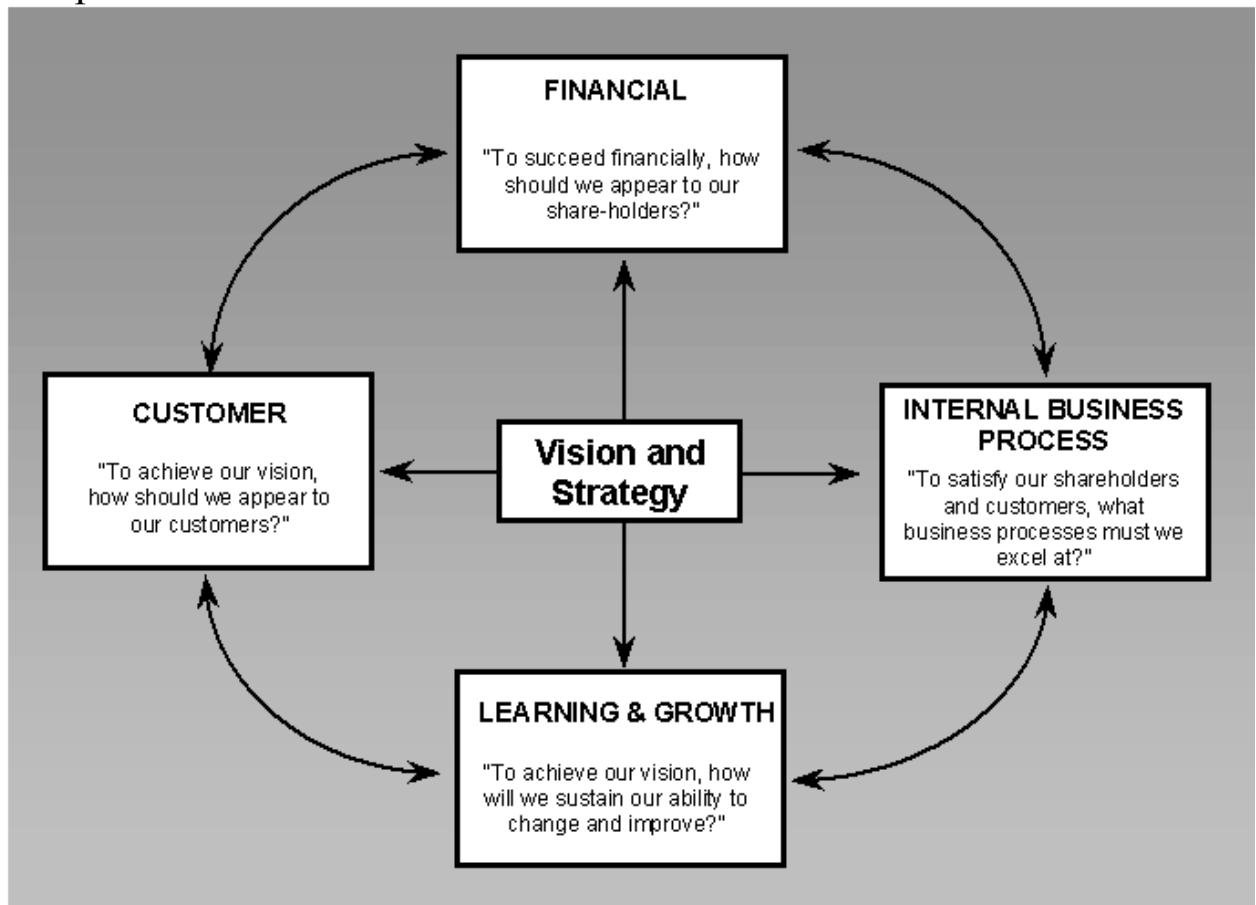
Why do it?

- To achieve strategic objectives.

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- To provide quality with fewer recourses.
- To eliminate non-value added efforts.
- To track progress.
- To align customer priorities.
- To evaluate process changes.
- To continually improves.

Perspectives of Balanced Scorecard



Measures:

- To determine how effectively and efficiently the process or service satisfies the customer.
- To identify improvement opportunities.
- To make decisions based on FACT and DATA.

Measurements should:

- Evaluate the quality of processes.
- Track our improvements.

- Focus our effort on our customers.
- Support our strategies.

Scorecard Potential Pitfalls

- Lack of well Defined Strategy
- Too much focus on the lagging measures
- Use of Generic Metrics
- Self-serving manager

Balanced Scorecard Importance

- Key performance helps to measure strategy at all levels of an organization.
- The methodology facilitates communication and understanding of business goals and strategies at all level of an organization.
- It enables executives to truly execute their strategy by identifying what should be done and measured.

2. Cybercrime



By Anil Pudasaini

What Is Cyber Crime?

Cybercrime is crime that involves a computer and a network. The computer may have been used in the commission of a crime, or it may be the target



Types of Cyber Crime

Fraud	Holiday Fraud
Identity Theft	Dating Fraud
Phishing Scams	Bullying
Viruses	Pension Fraud
Revenge Porn	Hacking
Online Hate Crime	Online Extremism
Grooming	Child Sexual
Stalking	Exploitation

...and this is just the tip of the iceberg

Phishing- The Warning Signs

The offer seems too good to be true

Links do not go to the place you would expect them to

Spelling mistakes throughout the email or in the subject line

The sender's address is not a genuine

There is a sense of urgency to respond to the email's request

Be aware for calls & texts too!

Viruses

- ▶ More than 317 million new pieces of malware -- computer viruses or other malicious software -- were created last year. That means nearly one million new threats were released each day. (As of 2015 reported by cnn.com)
- ▶ Malware can have harmful effects such as:
 - displaying irritating messages
 - stealing data
 - giving hackers control over your computer

14 'Blackhole' Exploit Kit Author Gets 7 Years

A Moscow court this week convicted and sentenced seven hackers for breaking into countless online bank accounts — including “**Paunch**,” the nickname used by the author of the infamous “**Blackhole**” exploit kit. Once an extremely popular crimeware-as-a-service offering, Blackhole was for several years responsible for a large percentage of malware infections and stolen banking credentials, and likely contributed to tens of millions of dollars stolen from small to mid-sized businesses over several years.

According to Russia's **ITAR-TASS news network**, **Dmitry “Paunch” Fedotov** was sentenced on April 12 to seven years in a Russian penal colony. In October 2013, the then 27-year-old Fedotov was arrested along with an entire team of other cybercriminals who worked to sell, develop and profit from Blackhole.

According to Russian security firm **Group-IB**, Paunch had more than 1,000 customers and was earning \$50,000 per month from his illegal



Blackhole v.1.0.0



Passwords

- ▶ Pitfall for a great amount of cyber security
- ▶ Most common passwords globally:
 1. 123456
 2. password
 3. 12345678
 4. qwertyuiop
 5. 12345
 6. 123456789
 7. football
 8. 1234
 9. 1234567
 10. Name123

If yours is one of these, please change it.

Top Tips For Online Safety

- ▶ Be careful of links and attachments in e-mails.



- ▶ Use different passwords for online accounts.



- ▶ Check websites security.



- ▶ If you are ever unsure about something online, do your research.



- ▶ Anti-virus software.



3. Page Rank

PAGE RANK

By Anish Shrestha
071/BCT/508

Page Rank : How it started

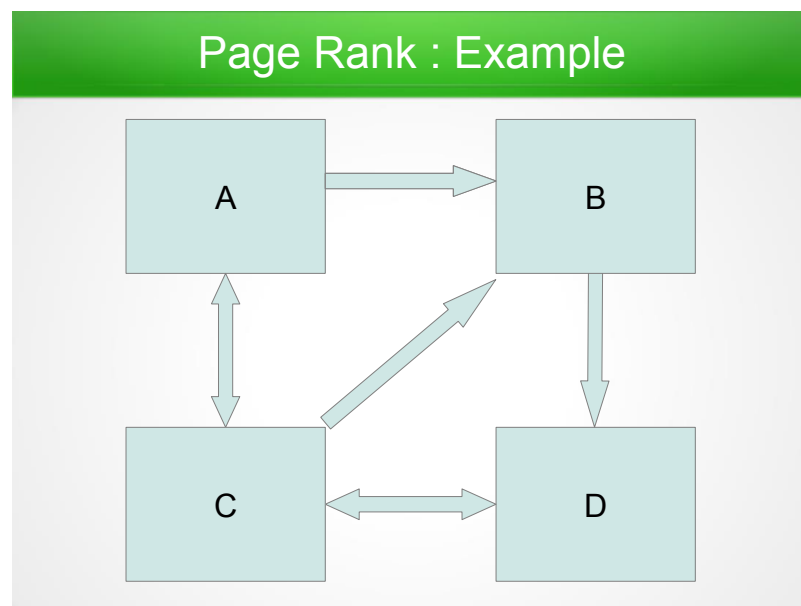
- Sir Tim Berners-Lee stored new websites in notebook
- Websites containing links to new websites
- As they grew search were added
- Hence Search Engines were created
- Problem:

How to sort the search result?

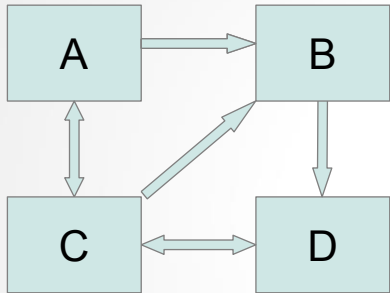
Page Rank : Formula

$$PR(A) = (1-d) + d(PR(T1)/C(T1) + \dots + PR(Ti)/C(Ti))$$

- $PR(A)$ is the PageRank of page A
- $PR(Ti)$ is the PageRank of pages Ti which link to page A
- $C(Ti)$ is the number of outbound links on page Ti
- d is a damping factor which can be set between 0 and 1.



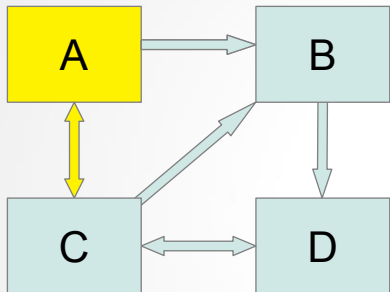
Page Rank : Example



	Iteration 0	Iteration 1	Iteration 2	Page Rank
A	1/4			
B	1/4			
C	1/4			
D	1/4			

Initial Page Rank of all Pages =
 $1/(\text{Total number of page})$

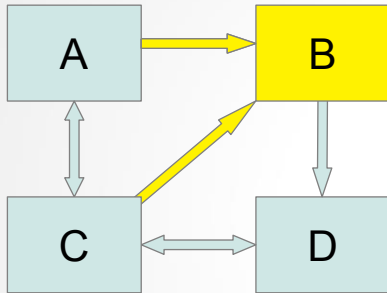
Page Rank : Example



	Iteration 0	Iteration 1	Iteration 2	Page Rank
A	1/4	1/12		
B	1/4			
C	1/4			
D	1/4			

Let $d = 1$
 $PR(A) = (1-d) + d(PR(C)/C(C))$
 $PR(A) = (1-1) + 1((1/4)/3)$

Page Rank : Example



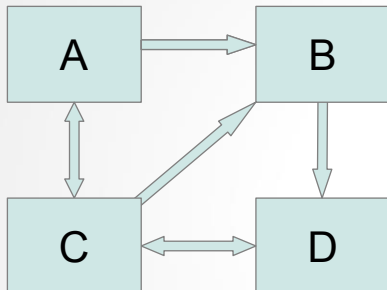
	Iteration 0	Iteration 1	Iteration 2	Page Rank
A	1/4	1/12		
B	1/4	2.5/12		
C	1/4			
D	1/4			

$$PR(B) = (1-d) + d (PR(A)/C(A) + PR(C)/C(C))$$

$$PR(B) = 0 + (1/4)/2 + (1/4)/3$$

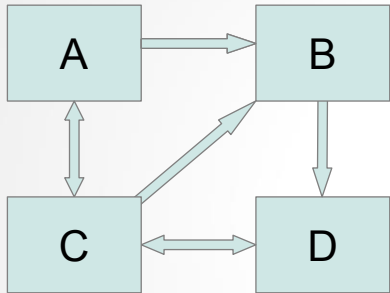
$$PR(B) = 2.5/12$$

Page Rank : Example



	Iteration 0	Iteration 1	Iteration 2	Page Rank
A	1/4	1/12		
B	1/4	2.5/12		
C	1/4	4.5/12		
D	1/4	4/12		

Page Rank : Example

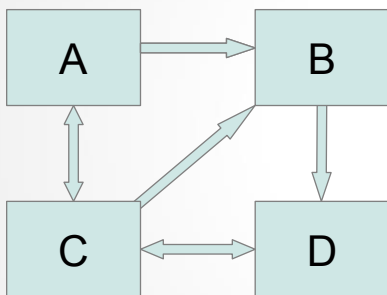


	Iteration 0	Iteration 1	Iteration 2	Page Rank
A	1/4	1/12	1.5/12 = 0.125	4
B	1/4	2.5/12	2/12 = 0.1667	3
C	1/4	4.5/12	4.5/12 = 0.375	1
D	1/4	4/12	4/12 = 0.3333	2

According to Google about 100 iteration is required for PageRank Algorithm.

Page Rank : Optimization

- Initial Page Rank as 1
- Using power method(Matrix)



$$\begin{array}{c}
 \mathbf{H} \\
 \left| \begin{array}{cccc} 0 & 0 & 1/3 & 0 \\ 1/2 & 0 & 1/3 & 0 \\ 1/2 & 0 & 0 & 1 \\ 0 & 1 & 1/3 & 0 \end{array} \right| \times \left| \begin{array}{c} \mathbf{V} \\ 1 \\ 1 \\ 1 \\ 1 \end{array} \right| = \left| \begin{array}{c} \mathbf{V}_1 \\ 1/3 \\ 5/6 \\ 1/2 \\ 1/3 \end{array} \right|
 \end{array}$$

$$\mathbf{V}_1 = \mathbf{H}\mathbf{V}$$

$$\mathbf{V}_2 = \mathbf{H}\mathbf{V}_1 = \mathbf{H}(\mathbf{H}\mathbf{V}) = \mathbf{H}^2\mathbf{V}$$

$$\mathbf{V}_n = \mathbf{H}^n\mathbf{V}$$

References : IOE notes and various sites

This note should be taken as reference and may be not enough for exam.

ALL THE BEST

BY: ER. ANKU JAISWAL